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Abstract

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Contents

1 Introduction

The rapid global emergence of antimicrobial resistance (AMR) represents one of the most critical threats that modern medicine is facing. As conventional antibiotics lose effiency against multi drug resistant bacterial pathogens, alternative therapeutic strategies have gained widespread attention. Among these is phage therapy, it is the targeted clinical application of bacteriophages (viruses that specifically infect and destroy bacteria) and has emerged as a highly promising solution [1].

Bacteriophages are the most abundant biological entities in the global biosphere, with an estimated population exceeding 10^{31} particles [2]. One of the main determinants of a phage's therapeutic viability is its host range (the specific spectrum of bacterial strains it can successfully infect). Molecular recognition is governed by highly specialized interactions between viral structural proteins (such as tail fibers, lysins etc.) and bacterial cell-wall receptors, therefore phages are extremely host specific. While this specificity is a major clinical asset that allows therapeutic phages to eradicate targeted pathogens without harming the patient, it also introduces a diagnostic challenge, that is matching a bacterial infection with an effective phage.

Traditional laboratory screening methods, such as plaque assays, are labor intensive, costly, and scale poorly against the millions of uncharacterized viral genomes generated by modern high throughput metagenomic sequencing. This neccessiates the development of fast and scalable computational tools, that are capable of predicting host ranges directly from viral genomic and protein-level representations [3].

However, state of the art computational prediction tools suffer from a fundamental limitation. They produce point predictions or uncalibrated scalar scores in $[0, 1]$ without rigorous, finite sample uncertainty quantification. In high stakes medical applications, point predictions offer no guarantee of reliability. A confidence score of 0.85 reflects a relative ranking not a calibrated conditional probability $\mathbb{P}(Y = y \mid X = x)$, and no existing method provides a finite sample guarantee that the true host is contained within a predicted candidate set. When a false negative means missing a life-saving treatment, such predictions are clinically insufficient

The main objective of this work is to resolve this limitation by contructing an uncertainty-aware statistical framework for phage-host prediction using conformal prediction [4], [5]. Conformal prediction is a distribution free method that transforms point scoring models into set valued predictions $C(X) \subseteq \mathcal{H}$. More importantly, these prediction sets are mathematically guaranteed to contain the true host label with a user specified confidence level $1 - \alpha$ (e.g., 90%), which holds strictly in finite samples under just the assumption of data exchangeability.

In this work we build on the Conformal Set Aggregation (CSA) framework [6]. We address four major challenges that arise when applying CSA to biological classification datasets:

(i) **Non-convex geometry in classification space:** We demonstrate why traditional convex envelopes, developed primarily for continuous regression spaces, are inefficient in classification settings. Here class conditional score vectors occupy disjoint, bounded clusters in $[0, 1]^K$. So, we propose non-convex (Radial and Strip) and Collapsed 1D envelope geometries that adapt to cluster topologies, and eliminate unpopulated empty space, which helps us reduce prediction set sizes.

(ii) **Closed form analytical calibration (τ):** We derive exact, closed form analytical expressions for the per point inclusion scaling function $\tau(\mathbf{s})$ across all envelope geometries. This eliminates the computationally expensive numerical binary search required by standard CSA, which reduces the calibration time complexity from $O(|S_2| \cdot N_{\text{iter}})$ to a single forward pass and $O(|S_2| \log |S_2|)$ sort.

(iii) **Class Conditional Validity:** Global calibration can yield under coverage for complex host classes and over coverage for simpler ones. To prevent this we incorporate per class quantile calibration ($\hat{t}_c$). This guarantees exact $1-\alpha$ coverage conditionally for every host label individually rather than just on average.

We validate our methodology systematically, we evaluated our framework on two tasks of increasing complexity. First, a controlled binary Gram-type classification task ($N = 2$ classes) to validate the geometric envelope constructions, and then on a multi class host-level prediction task ($N = 54$ bacterial genera) using real-world protein language model embeddings.

The remainder of this report is organized as follows: Chapter 2 reviews the biological background of bacteriophages, computational state-of-the-art methods, and the mathematical foundations of split conformal prediction and CSA. Chapter 3 formalizes the prediction tasks, develops the non-convex envelopes, the nonconformity transformation, and the analytical derivation of $\tau(\mathbf{s})$. Chapter 4 details the dataset architecture, protein to phage NaN aware mean pooling operators, and cross split protein cluster contamination filtering. Chapter 5 presents synthetic experiments and empirical evaluation results for both gram-type and host-level classification tasks. Finally, chapter 6 summarises our findings, discusses the biological limitations, and outlines directions for future research.

2 Background & State of the Art

In this section, we provide the biological, computational and mathematical background and context for the methods developed in this report. We will begin by introducing bacteriophages and the host-prediction problem in Section 2.1, which motivates the need for computational approaches to find a solution. We will then go over the already existing and state-of-the-art computational techniques in Section 2.2, progressing from the classical sequence similarity techniques to the much nore efficient and modern protein embeddings techniques. We will also look at the limitations that all the currently exisiting methods have in common, for example the absence of rigourous distribution-free uncertainity quantification. In the remaining part of the section, 2.3 and 2.4 we will introduce the mathematical framework of conformal prediction and the score-based aggregation method of in [6], on which we will build our work.

2.1 Bacteriophages and Host-Prediction Problem

Bacteriophages, commonly referred to as phages, are viruses that specifically infect and replicate within bacteria. They are one of the most abundant biological entities on Earth with an estimated population of 10^{31} in the global biosphere [2], and therefore play a very important role in shaping the microbial communities and evolution. In the recent years the interest in bacteriophages has increased due to the growing threat of antimicrobial resistance and the development of phage therapy as an alternate treatment method to the traditional antibiotics [1].

One of the key characteristics of of phages is its host range, i.e. the set of bacterial hosts that the phage is capable of infecting. The host range of a phage is not a fixed property, it is determined by molecular interactions between the phage and the bacterial surface structures during the cycle of infection [2], [7]. Each phage attaches itself to the bacterial host, injects its viral genetic material which are able to replicate inside the host. Therefore, some of the phages only infect a few hosts (have a small host range) as they can only infect a limited number of bacterial strains while there are other which are capable of infecting a broad range of bacterias.

Identifying which phage infects which host, can have several applications both in environmental and clinical studies. In clinical, the host range helps us determine how suitable a phage is for designing the treatment for phage therapy (an alternate method to antibiotics). In environmental studies it is valuable to know the host being infected to understand the ecological interactions between viruses and microbial communities [7] and evolution in the biosphere.

However, host range has also brings its own obstacles. In phage therapy, we need to find precisely the phage that infects the given bacterial pathogen which is the cause of the infection, and for that we need a fast reliable host prediction method. Traditionally, the method of determining the host range used to depend on laboratory experiments in which candidate phages were tested aginst a number of bacterial hosts; although this method was able to give accurate reliable results but it was extremely expensive and time consuming. Given the rapid increase in the phage genomes generated by modern technology, the number of possible phage-host interacions has increased astronomically, and this leads us to the need of having a compuational prediction method. We need a way which can complement the lab experiment by providing with a shorter list of candidate phages before we move on to expermiental validation [3] or even directly infer the host information directly from a given phage.

We can formulate this host prediction problem as follows: let $\mathcal{X} \subseteq \mathbb{R}^d$ be the feature space representing phage representations, and let $\mathcal{H} = \{h_1, \cdots, h_N\}$ be the finite label space of possible bacterial hosts.

In a single-host classification setting, we observe random pairs $(X, H) \sim \mathbb{P}_{X,H}$, where $X \in \mathcal{X}$ is a feature vector representing a phage and $H \in \mathcal{H}$ represents the true host. In a general multi-host setting, the target is a non-empty subset of hosts $H^* \subseteq \mathcal{H}$, i.e., $H^* \in \mathcal{P}(\mathcal{H}) \setminus \{\emptyset\}$, where $\mathcal{P}(\mathcal{H}) = 2^{\mathcal{H}}$ denotes the power set of $\mathcal{H}$ 3.1.

Our primary objective is to construct a set-valued prediction mapping:

$$C : \mathcal{X} \longrightarrow \mathcal{P}(\mathcal{H})$$

which maps an unlabelled test phage $X_{n+1} \in \mathcal{X}$ to a prediction set $C(X_{n+1}) \subseteq \mathcal{H}$ such that it guarantees that the true host is included in the prediction set with a with a user specified statistical confidence, $1 - \alpha \in (0, 1)$:

$$\mathbb{P}\left(H_{n+1} \in C(X_{n+1})\right) \geq 1 - \alpha$$

where $\alpha \in (0, 1)$ denotes the target error probability. this

2.2 Current Computational Approaches and Limitations

The problem of predicting which phage infects which host bacteria has been studied computationally for over a decade and the methods have significantly evolved during this time. Prediction methods have progressed from simple rule based alignemnt to deep representation learning. Here, we will summarize each the primary methodologies of these methods and along with the common limitations they share that motivate us for this work.

2.2.1 Sequence Similarity and Protein Content Methods

The early host prediction tools rely on sequence alignment between a phage and bacterial genomes, assuming that similarity between the two indicates infection compatibility. HostPhinder [8] represents each genome g by its normalised k-mer frequency vector $\mathbf{v}(g) \in \mathbb{R}^{4^k}$ and assigns the predicted host through nearest-neighbour search over a reference database $\mathcal{D}_{\text{ref}} = \{(g_i, y_i)\}_{i=1}^{N}$:

$$\hat{y} = \underset{y \in \mathcal{H}}{\operatorname{argmax}} \max_{g_i : y_i = y} \operatorname{sim}(\mathbf{v}(g_{\text{test}}), \mathbf{v}(g_i)) .$$

WIsH [9] uses a probabilistic approach, it models each candidate host genome as a variable- order Markov chain with parameters θ_h and then selects the host that maximises the log-likelihood of the new test phage sequence $S = (a_1, \ldots, a_L)$:

$$\hat{y} = \underset{h \in \mathcal{H}}{\operatorname{argmax}} \ \ell(S \mid \theta_h), \qquad \ell(S \mid \theta_h) = \sum_{i=k+1}^{L} \log \mathbb{P}(a_i \mid a_{i-k}, \ldots, a_{i-1}; \theta_h).$$

The protein-content methods overcome the limitation of whole-genome alignment by decomposing the phage into its constituent proteins. RaFAH [10] clusters phage proteins into M protein clusters via MMseqs2 [11], then it represents each phage by a binary membership vector $\mathbf{x} \in \{0, 1\}^M$, and trains a random forest to map this profile to a distribution over host genera.

$$f_{\text{RF}} : \{0, 1\}^M \longrightarrow \Delta^{|\mathcal{H}|-1}, \quad \text{where } \hat{\mathbf{p}} = f_{\text{RF}}(\mathbf{x}), \quad \sum_{y \in \mathcal{H}} \hat{p}_y = 1$$

This representation is directly relevant to the present work, as our pipeline operates on a related protein-cluster construction, and the data contamination problem discussed in 4.3 arises from the same source.

2.2.2 Protein Language Models and Embedding-Based Methods

The current state of the art techniques use protein language models (pLMs), which are neural architectures trained on millions of protein sequences through self-supervised learning [12], [13]. Given a protein sequence $A = (a_1, \ldots, a_n) \in \mathcal{A}^n$ over the amino acid alphabet $\mathcal{A}$ of size 20, an encoder Φ_{pLM} produces a residue-level embedding matrix $\mathbf{Z} = \Phi_{\text{pLM}}(A) \in \mathbb{R}^{n \times d}$, it is then mean-pooled to a fixed-length sequence representation:

$$\mathbf{z} = \frac{1}{n} \sum_{i=1}^{n} \mathbf{Z}_{i,:} \ \in \ \mathbb{R}^d.$$

The embedding $\mathbf{z}$ encodes biochemical, structural, and evolutionary information without requiring explicit sequence alignment. Building on ProtT5 [12], tools such as EMPATHI [14] and SUBLYME [15] use these representations for phage protein functional annotation and metagenomic lysin discovery respectively. These tools are directly involved in this work: the K-dimensional score vector $\mathbf{s} = (s_1(X), \ldots, s_K(X))^T \in [0,1]^K$ that serves as input to our conformal framework is produced by a pipeline built on these annotations, the precise construction of which is described in 4. For host prediction, protein embeddings have shown substantially improved accuracy over sequence- similarity baselines [16], [17]. So this suggests that the embedding space contains sufficient information for reliable host assignment.

2.2.3 Shared Limitation: Absence of Uncertainty Quantification

Despite their empirical success of all the methods seen, they all share a fundamental limitation: they produce point predictions (a single host label $\hat{y}$ or a scalar score $\hat{s}(x, y) \in [0, 1]$) without any formal uncertainty quantification.

Uncertainty Quantification (UQ) is defined as the mathematical framework used to rigorously measure, propagate, and bound prediction errors. A raw scalar output $\hat{s}(x, y) = 0.87$ merely reflects an uncalibrated relative ranking statistic, not a true conditional probability $\mathbb{P}(Y = y \mid X = x)$. Existing host prediction tools fail to provide UQ because no tool offers a finite sample statistical guarantee that the true bacterial host is contained within a candidate prediction set with a predefined target probability $1 - \alpha$.

Existing UQ frameworks only offer partial solutions. For example, the Bayesian approaches require strong assumptions on the distribution distributional that are difficult to justify for biological score data, while post-hoc calibration methods such as Platt scaling provide no finite-sample guarantees on unseen test data [5].

These limitations are consequential in phage therapy, where an incorrectly predicted host can directly harm a patient. This motivates the adoption of conformal prediction [4]. This is a distribution-free framework that wraps any existing scoring model in a statistical construction, and produces prediction sets $C(X) \subseteq \mathcal{H}$ with exact finite-sample coverage guarantees while only requiring exchangeability of calibration and test data.

2.3 Conformal Prediction

Conformal prediction, initially developed by Vovk, Gammerman and Shafer [4] and then later made accessible by Angelopoulos and Bates [5], is a framework for constructing prediction sets that have a coverage guarantee. Unlike the methods seen in 2.2 it can provide us with a set of possible outcomes (not a point prediction), have a mathematical guarantee and yet requires no distributional assumptions on the data, except for the exchangibility condition. Here, we will formalize the framework of conformal prediction and analyze its finite-sample coverage properties.

Definition 2.1 (Exchangeability)**.**
A finite sequence of random variables $Z_1, Z_2, \dots, Z_m$ taking values in a space $\mathcal{Z}$ is exchangeable if, for every permutation σ of the indices $\{1, 2, \dots, m\}$, the joint distribution of the permuted sequence is identical to that of the original sequence:

$$(Z_1, Z_2, \dots, Z_m) \stackrel{d}{=} (Z_{\sigma(1)}, Z_{\sigma(2)}, \dots, Z_{\sigma(m)})$$

where $\stackrel{d}{=}$ denotes equality in distribution.
Or, equivalently for all measurable sets $A_1, A_2, \dots, A_m \subseteq \mathcal{Z}$ and all permutations $\sigma \in S_m$:

$$\mathbb{P}(Z_1 \in A_1, Z_2 \in A_2, \dots, Z_m \in A_m) = \mathbb{P}(Z_{\sigma(1)} \in A_1, Z_{\sigma(2)} \in A_2, \dots, Z_{\sigma(m)} \in A_m)$$

Independent and identically distributed (i.i.d.) seqeunces are always exchangeable, but the converse does not always hold. Thus, exchangeability is considered to be a weaker condition than i.i.d., since it requires that the observations have no special ordering. This condition can easily be met by splitting any dataset randomly [4], [6].

2.3.1 Mathematical Setup and Prediction Set

Let $(X, H) \in \mathcal{X} \times \mathcal{H}$ be a random input-label pair, where $\mathcal{X}$ is the feature space that represents our object of interest (in our case it is the phage protein level score vectors) and $\mathcal{H} = \{h_1, \cdots, h_N\}$ is the discrete label space (set of candidate hosts).

We assume that we have a calibration dataset, $D_{\text{cal}} = \{(X_1, H_1), \cdots, (X_n, H_n)\}$, of n exchangable data points drawn from an unknown joint distribution, $P_{X,H}$. Then, for a new unlabelled test input, X_{n+1} our aim is to to construct a prediction set function $C : \mathcal{X} \to 2^{\mathcal{H}}$ dependening on the data which maps the test input to a subset of candidate labels, i.e., $C(X_{n+1}) \subseteq \mathcal{H}$, such that the true label falls in in the set with a defined coverage that is:

$$\mathbb{P}(H_{n+1} \in C(X_{n+1})) \geq 1 - \alpha$$

where $\alpha \in (0, 1)$ is the significance level. In our experiments we set $\alpha = 0.1$ so that the guarantee is 90%.

Unlike point predictors, $C(X)$ dynamically reflects model uncertainty. It returns a singleton $|C(X)| = 1$ when confident, a multi-label set $|C(X)| \geq 2$ when uncertain between candidates, or an empty set $|C(X)| = 0$ for out-of-distribution inputs. The efficiency of the model can be quantified by the expected set size $\mathbb{E}[|C(X)|]$, an optimal predictor should acheive a coverage $\geq 1 - \alpha$ with the smallest possible set size.

2.3.2 Split conformal prediction

In split conformal prediction, the available data is divided into a training set and a disjoint calibration set, $D_{\text{cal}} = \{(X_i, H_i)\}_{i=1}^n$. The training set is used to fit an underlying scoring model $\hat{f} : \mathcal{X} \times \mathcal{H} \to \mathbb{R}$. We then define a nonconformity score function $s : \mathcal{X} \times \mathcal{H} \to \mathbb{R}$ based on $\hat{f}$. This score function quantifies how unusual a pair (x, h) is relative to the learned patterns, smaller values of $s(x, y)$ indicate stronger conformity between x and label h

Then using the calibration set , $\{(X_i, H_i)\}_{i=1}^n$, we evaluate the nonconformity scores:

$$S_i = S(X_i, H_i), \quad \forall i = 1, \cdots, n$$

To guranatee that out error rate is less than α we need to compute the conformal calibration qunatile, $\hat{q}$, from these calibration scores

Definition 2.2 (Conformal Calibration Quantile)**.**
The conformal calibration quantile q_α is defined as the empirical $\frac{\lceil (n+1)(1-\alpha) \rceil}{n}$-th quantile of calibration scores $\{S_1, \ldots, S_n\}$:

$$q_\alpha := \text{Quantile}\left(\{S_1, \ldots, S_n\}\,;\, \frac{\lceil (n+1)(1-\alpha) \rceil}{n}\right)$$

Equivalently, expressed using the empirical cumulative distribution function:

$$q_\alpha = \inf \left\{ q \in \mathbb{R} : \frac{1}{n} \sum_{i=1}^{n} \mathbf{1}(S_i \leq q) \geq \frac{\lceil (n+1)(1-\alpha) \rceil}{n} \right\}$$

Then, for a new input, X_{n+1} the conformal prediction set, $C(X_{n+1})$, is constructed by collecting all the candidate labels in $H \in \mathcal{H}$ whose nonconformity scores do not exceed q_α:

$$C(X_{n+1}) = \left\{ H \in \mathcal{H} \;\middle|\; s(X_{n+1}, H) \leq q_\alpha \right\} \tag{2.1}$$

The validity of this construction follows from exchangeability. Let $S_{(1)} \leq \cdots \leq S_{(n)}$ denote the order statistics of the calibration scores $\{S_1, \ldots, S_n\}$, and augment this sequence with $S_{(n+1)} = +\infty$. Then $q_\alpha = S_{(\lceil (n+1)(1-\alpha) \rceil)}$ is the $\lceil (n+1)(1-\alpha) \rceil$-th order statistic of the augmented set $\{S_1, \ldots, S_n, S_{n+1}\}$. Since $(Z_1, \ldots, Z_{n+1})$ are exchangeable, the rank of S_{n+1} among the $n+1$ augmented scores is uniformly distributed on $\{1, \ldots, n+1\}$. Consequently:

$$\mathbb{P}\big(S_{n+1} \leq q_\alpha\big) = \mathbb{P}(\text{rank}(S_{n+1}) \leq \lceil (n+1)(1-\alpha) \rceil) = \frac{\lceil (n+1)(1-\alpha) \rceil}{n+1} \geq 1 - \alpha.$$

Theorem 2.3 (Marginal Coverage Guarantee [4], [18])**.** *If* $(X_1, H_1), \ldots, (X_{n+1}, H_{n+1})$ *are exchangeable, then the prediction set* $C(X_{n+1})$ *defined in Equation 2.1 satisfies:*

$$\mathbb{P}\left(H_{n+1} \in C(X_{n+1})\right) \geq 1 - \alpha$$

Furthermore, if calibration scores $S_1, \ldots, S_n$ *are almost surely distinct continuous random variables, the coverage bound is tight:*

$$1 - \alpha \leq \mathbb{P}\left(H_{n+1} \in C(X_{n+1})\right) \leq 1 - \alpha + \frac{1}{n+1} \tag{2.2}$$

Crucially, theorem 2.3 holds strictly in finite samples without requiring parametric assumptions on $s(\cdot, \cdot)$ or $\mathbb{P}_{X,H}$. While the choice of scoring function s governs efficiency (set size), valid statistical coverage is guaranteed by construction in a single computational pass over D_{cal} [6].

2.3.3 Marginal & Conditional Coverage

Standard conformal prediction satisfies marginal coverage, which means that the $(1-\alpha)\%$ coverage guarantee holds on average across all possible choices of the calibration set, D_{cal}, and test samples, (X_{n+1}, H_{n+1}):

$$\mathbb{E}_{D_{cal},(X_{n+1},H_{n+1})}\left[\mathbb{I}(H_{n+1} \in C(X_{n+1}))\right] = \mathbb{P}(H_{n+1} \in C(X_{n+1})) \geq 1 - \alpha$$

However, the marginal coverage does not ensure the conditional coverage, given specific features $(X = x)$, or specific classes $(H = h)$. So, for a dataset that has class imbalance, which is very common in phage-host interaction, a marginal model can satisfy the coverage guarantee by potentially over-covering a majority class (100%) and under- covering a small class (0%) [6], [19]. For instance, if $\mathcal{H} = \{-1, 1\}$ with class proportion $\mathbb{P}(H = -1) = 0.9$, a predictor acheiving 100% coverage on $H = -1$ and 0% coverage on $Y = 1$ still attains an overall 90%. In phage-host classification, such conditional under-coverage on minority host may be a problem.

While feature-conditional coverage ($\mathbb{P}(H \in C(X) \mid X = x) \geq 1 - \alpha$) is mathematically impossible to achieve in finite samples without strong parametric assumptions [4], [6], class-conditional coverage (Mondrian conformal prediction) is exact and valid in finite samples. We partition D_{cal} into N disjoint subsets $D_{\text{cal}}^{(h)} = \{(X_i, H_i) \in D_{\text{cal}} : H_i = h\}$ and evaluate separate class-specific quantiles $q_\alpha^{(h)}$, enforcing exact coverage per class:

$$\mathbb{P}\left(H_{n+1} \in C(X_{n+1}) \,\middle|\, H_{n+1} = h\right) \geq 1 - \alpha, \quad \forall h \in \mathcal{H} \tag{2.3}$$

Because exchangeability holds within each class-conditional partition $D_{\text{cal}}^{(h)}$, evaluating per-class quantiles preserves exact finite-sample validity across every candidate host, as formalised in Section 3.5.2.

2.4 Conformal Set Aggregation (CSA)

The standard split conformal framework in 2.3.2 takes a scalar non-conformity score $S(x, h) \in \mathbb{R}$. In this report, the prediction pipeline produces a K-dimensional score vector of scores $s \in [0, 1]^K$ for each phage, where K is the number of distinct functional categories. We may lose the geometric structure of the score distribution if we aggregate this vector to a scalar value before applying conformal prediction. For such settings, Ochoa Rivera, Patel and Tewari [6] have proposed a score-based aggregation (CSA) method for conformal prediction.

2.4.1 Ensemble Score Vectors

Let $\mathcal{S} \subseteq [0, 1]^K$ be the score space and let $s = (s_1, \cdots, s_K) \in \mathcal{S}$ be the vector of scores produces by the K models, for each phage. Now, the CSA method treats the entire vector as the object of inference rather than reducing it into a single value. The main idea is to define a region, called an envelope, $E \subseteq [0, 1]^K$. This envelope consists of the region where the score vectors of the true hosts for that particular phage are concentrated.

Mathematically, let $\{\phi_h\}_{h \in \mathcal{H}}$ be a family of label-dependent maps $\phi_h : [0, 1]^K \to [0, 1]^K$ such that, when h is the true label, the transformed score vector $\phi_h(\mathbf{s})$ concentrates near the origin in $\mathbb{R}_+^K$. The CSA method constructs an envelope $E \subseteq \mathbb{R}_+^K$ as the intersection of M half-spaces induced by unit directions $\{\mathbf{u}_m\}_{m=1}^M$ and quantile thresholds $\{\tilde{q}_m\}_{m=1}^M$:

$$E = \bigcap_{m=1}^{M} \left\{ \mathbf{s} \in \mathbb{R}_+^K : \langle \mathbf{s}, \mathbf{u}_m \rangle \leq \tilde{q}_m \right\}. \tag{2.4}$$

By construction, E is convex. Membership of a transformed score in the scaled envelope $t \cdot E$ is tested coordinate-wise along each direction. The exact transformation ϕ_h used in this work is formalized in Section 3.3.

2.4.2 The CSA Framework

The core innovation of the CSA method is in the calibration process. It splits the calibration data into two subsets which have different roles. This is necessary for the exchangeability argument that provides us with the coverage guarantee. If we used the same calibration points to construct both, we would get overfitting which would introduce bias which would invalidate the distruibution free coverage guarantee [6].

(i) **Shape Discovery Set (S_1):** Used to model the spatial distribution of scores and learn the geometric shape, orientation, and boundaries of the envelope $E \subset \mathbb{R}^K$.

(ii) **Size Scaling Set (S_2):** Used to determine the scalar scaling parameter $\hat{t} \in \mathbb{R}^+$. $\hat{t}$ inflates or deflates E until exact empirical coverage $1 - \alpha$ is attained over S_2:

$$\hat{t} = \inf \left\{ t > 0 \;\middle|\; \frac{1}{|S_2|} \sum_{\mathbf{s} \in S_2} \mathbf{1}\left(\phi_H(\mathbf{s}) \in t \cdot E\right) \geq \frac{\lceil (|S_2| + 1)(1 - \alpha) \rceil}{|S_2|} \right\}$$

This $\hat{t}$ is the smallest scaling factor such that the scaled envelope $\hat{t} \cdot E$ achieves the required $1 - \alpha$ coverage. The scaled envelope is defined linearly as:

$$\hat{t} \cdot E = \{t \cdot x : x \in E\}, \quad \hat{t} > 0$$

In the original CSA paper [6], finding $\hat{t}$ is presented as a root-finding problem solved through iterative numerical optimization or binary search over candidate scaling values t. At test time, a candidate label h is included in the prediction set $C(\mathbf{s}_{n+1})$ if and only if its transformed score vector falls inside the scaled envelope:

$$C(\mathbf{s}_{n+1}) = \{h \in \mathcal{H} : \phi_h(\mathbf{s}_{n+1}) \in \hat{t} \cdot E\}.$$

2.4.3 Limitations of CSA in Classification

This CSA framework was originally developed and analyzed in a regression setting [6], and has some major limitations when it is applied to classification problems:

(i) **Convexity of Envelope:** In [6], the conformal envelope E is constructed as an intersection of half-spaces and is therefore convex. But in classification, the score vectors for different classes concentrate in seperate regions in the $[0, 1]^K$ score space. Therefore, if we force a convex envelope across the regions, we will create a region that covers a large area of empty score space in between, which would inflate the volume of the envelope. This would lead to an over conservative and uninformative prediction sets, $|C(\mathbf{s})|$, as they would be really large.

(ii) **Calibration Inefficiency:** Originally the method uses an iterative numerical optimication or a binary search to find the optimal scaling factor, $\hat{t}$ over S_2. In cases when the score space is high dimensional, the iterative approach can be really expensive computationally.

(iii) **Only Marginal Guarantee:** The scaling calculations in the original method only provide us with a marginal guarantee. This can lead to poor performance on the biological datasets, which are imbalances

These limitations directly motivate our methodological contributions in 3. This includes designing non-convex envelopes 3.4, deriving closed-form analytical solutions for per-point τ-values to enable one-pass calibration 3.5, and integrating class conditional thresholds to guarantee reliable host-range predictions.

3 Methodology

In this section, we will develop the mathematical framework used to construct prediction sets with guaranteed finite-sample phage-host classification. Section 3.1 formalises the prediction task and addresses missing data in our score vector. Then, building on the limitations of the CSA method identified in Section 2.4.3, Section 3.2 explains why non-convex envelopes are better suited for classification, and Section 3.3 defines the label-dependent nonconformity transformation used throughout this report to center valid score distributions near the origin. Section 3.4 then presents the three envelope geometries evaluated in this work, Radial, Strip, and Collapsed 1D, and derives the calibration scaling factor $\hat{t}$.

3.1 Problem Formalization

We will now apply the general conformal prediction setup from Section 2.3 to our host prediction task. Recall from Section 2.1 that $\mathcal{H} = \{h_1, \ldots, h_N\}$ represents the discrete set of N candidate bacterial hosts. Our goal is to build a function $C : \mathcal{X} \to \mathcal{P}(\mathcal{H})$ that outputs a set of hosts while maintaining the coverage guarantee in Equation (2.1)

In our work, the feature space, $\mathcal{X}$, does not use the raw phage sequences directly. Instead it consists of fixed-length score vectors (K dimensional) generated by the protein annotation process introduced in Section 2.2.2. So each score s_k corresponds to a specific functional protein category (e.g., capsid, tail, lysin), where K is the total number of categories. A specific phage genome may lack proteins in different functional categories, therefore the score vector may have structural missingness:

$$\mathbf{s} = (s_1, \ldots, s_K)^T \in ([0,1] \cup \{\text{NA}\})^K.$$

Here, $s_k = \text{NA}$ is the structural absence, which means that for category k no annotated protein was identified in the phage sequence, rather than a zero-evidence score or missingness caused by random measurement noise. This value is missing by design, not due to noise, and hence we need to handle it explicitly rather than simply imputing it.

To retain the mathematical guarantee without imputations, we attach a missingness mask, $\mathbf{m}$, to every score vector. We represent each score observation as a paired vector $(\mathbf{s}, \mathbf{m}) \in ([0,1] \cup \{\text{NA}\})^K \times \{0,1\}^K$:

$$\mathbf{m} = (m_1, \ldots, m_K)^T \in \{0,1\}^K, \qquad m_k = \mathbf{1}(s_k \neq \text{NA}).$$

Given a calibration set $D_{\text{cal}} = \{(\mathbf{s}_i, \mathbf{m}_i, H_i)\}_{i=1}^n$ of exchangeable samples, and an unlabelled test sample $(\mathbf{s}_{n+1}, \mathbf{m}_{n+1})$ drawn exchangeably from an unknown joint probability distribution. Our goal is to construct a set-valued prediction mapping $C : ([0,1] \cup \{\text{NA}\})^K \times$

$\{0,1\}^K \to \mathcal{P}(\mathcal{H})$ at significance level $\alpha \in (0,1)$ such that:

$$\mathbb{P}\Big(H_{n+1} \in C(\mathbf{s}_{n+1}, \mathbf{m}_{n+1})\Big) \geq 1 - \alpha.$$

At the same time, to maximize the preditive efficiency, we want to minimize the expected set size $\mathbb{E}\left[|C(\mathbf{s}, \mathbf{m})|\right]$. The sets $C(\mathbf{s}_{n+1}, \mathbf{m}_{n+1})$ as defined above may be empty when no candidate label's transformed score falls within the calibrated envelope. The two prediction tasks in this report resolve this case differently 3.4.
The total number of observed functional dimensions for a given phage is given by the L_1 norm of the mask, $\|\mathbf{m}\|_1 = \sum_{k=1}^{K} m_k$. Then we have $\mathbf{s_m} \in [0,1]^{\|\mathbf{m}\|_1}$, the sub-vector restricted strictly to the observed dimensions where $m_k = 1$. All prediction functions and nonconformity scoring operators developed in this report are defined over the product space of pairs $(\mathbf{s}, \mathbf{m})$, rather than $\mathbf{s}$ alone. This ensures our nonconformity scores and envelope tests remain mathematically valid for any missingness pattern, not just fully observed data, $\mathbf{m} = \mathbf{1}$.

We apply this framework to two host prediction tasks:

(i) **Gram-type classification** ($N = 2$): $\mathcal{H} = \{\text{gram-neg}, \text{gram-pos}\}$, evaluated over K functional category dimensions.

(ii) **Host-level classification** ($N > 2$): $\mathcal{H}$ represents the full set of bacterial hosts in our dataset. For each host $h \in \mathcal{H}$, the score space is decomposed into host-specific sub-vectors of dimension K_h, and an individual envelope E_h is fitted independently on training phages with true host h.

In practice, host-level classification is not evaluated over the full host space $\mathcal{H}$ at test time. We first obtain a gram-type prediction set $C_{\text{gram}}(\mathbf{s}) \subseteq \{\text{gram-neg}, \text{gram-pos}\}$ from the procedure of Section 3.3.1, and use it to restrict the candidate host set to

$$\hat{\mathcal{H}}(\mathbf{s}) = \bigcup_{g \in C_{\text{gram}}(\mathbf{s})} \mathcal{H}_g \subseteq \mathcal{H},$$

where $\mathcal{H}_g \subset \mathcal{H}$ denotes the set of hosts of gram-type g. Let $s_h \in \big([0,1] \cup \{NA\}^{K_h}\big)$ denotes the restriction of s to the K_h score dimensions associated with host h. The host-level prediction map of this report is therefore the composition

$$C_{\text{host}}(\mathbf{s}, \mathbf{m}) = \big\{h \in \hat{\mathcal{H}}(\mathbf{s}) \bigm| \phi_h(\mathbf{s}_h) \in \hat{t}_h \cdot E_h\big\}, \tag{3.1}$$

rather than a single-stage test over all of $\mathcal{H}$. This cascade reduces the number of per-host envelope evaluations required at test time, at the cost of making C_{host} dependent on the upstream gram-type prediction.

The following result confirms that composing the gram type and host level stages does not degrade the coverage guarantee of Equation (3.1).

Proposition 3.1 (Cascade Validity)**.**
Under exchangeability, if gram type classification is class conditionally valid at level α and each candidate host envelope satisfies validity level α_h, the cascaded predictor satisfies

$$\mathbb{P}\big(H^\star \notin C_{host}(\mathbf{s})\big) \leq \alpha + \max_{h \in \mathcal{H}} \alpha_h. \tag{3.2}$$

Proof. The miscoverage event $\{h^\star \notin C_{\text{host}}(\mathbf{s})\}$ implies either that the true host was pruned by the Gram type filter ($h^\star \notin \mathcal{H}_{\text{cand}}(\mathbf{s})$, which requires $g^\star \notin C_{\text{gram}}(\mathbf{s})$) or that $h^\star$ was admitted to the candidate pool but rejected by its host envelope ($\tau_{h^\star}(\mathbf{s}) > \hat{t}_{h^\star}$). Applying the union bound:

$$\begin{aligned}
\mathbb{P}\big(h^\star \notin C_{\text{host}}(\mathbf{s})\big) &\leq \mathbb{P}\big(g^\star \notin C_{\text{gram}}(\mathbf{s})\big) + \mathbb{P}\big(\tau_{h^\star}(\mathbf{s}) > \hat{t}_{h^\star}\big) \\
&= \sum_g \mathbb{P}(G = g)\,\mathbb{P}(g^\star \notin C_{\text{gram}} \mid G = g) + \sum_h \mathbb{P}(H = h)\,\mathbb{P}(\tau_h > \hat{t}_h \mid H = h) \\
&\leq \alpha + \max_{h \in \mathcal{H}} \alpha_h.
\end{aligned}$$

□

For $\alpha = 0.10$, this gives a conservative worst-case coverage lower bound of $1 - 2\alpha = 0.80$; Section 5.3 reports the much tighter empirical coverage actually achieved.

3.2 Non-Convex Envelopes in Classification

Section 2.4.3 identified envelope convexity as one of the main limitations of standard CSA method [6] when applied to classification. While convex envelopes naturally model continuous target spaces in regression, classification score vectors concentrate in disjoint, class-specific clusters across the score space. Therefore, it leads to the envelope being streched over regions of empty spaces, making the prediction sets unnecessarily large. Here, we wil describe the alternate method that we have used in this report.

Formally, let $\mathcal{C}_h \subseteq \mathbb{R}_+^K$ denote the support of the transformed NC vector $\phi_h(\mathbf{s})$ for true label $H = h \in \mathcal{H}$. In multi-class classification, these supports are strictly pairwise separated:

$$\mathcal{C}_h \cap \mathcal{C}_{h'} = \emptyset, \qquad \operatorname{dist}(\mathcal{C}_h, \mathcal{C}_{h'}) > 0, \qquad \forall h \neq h' \in \mathcal{H}. \tag{3.3}$$

A single convex envelope $E_{\text{conv}} = \operatorname{conv}\left(\bigcup_{h \in \mathcal{H}} \mathcal{C}_h\right)$ must contain every line segment connecting points in $\mathcal{C}_h$ and $\mathcal{C}_{h'}$. These line segments inevitably traverse unpopulated space. Let us define the empty gap as gap $:= E_{\text{conv}} \setminus \bigcup_{h \in \mathcal{H}} \mathcal{C}_h$, the Lebesgue volume decomposes as:

$$\operatorname{Vol}(E_{\text{conv}}) = \sum_{h \in \mathcal{H}} \operatorname{Vol}(\mathcal{C}_h) + \operatorname{Vol}(\text{gap}), \qquad \text{where } \operatorname{Vol}(\text{gap}) > 0. \tag{3.4}$$

Test points falling in this gap are included in the prediction set by construction, $|C(\mathbf{s})|$, and they make them larger without improving coverage. This inefficiency scales with the number of classes $N = |\mathcal{H}|$, making shared convex hulls particularly suboptimal for hostlevel classification with multiple classes ($N > 2$).

Also very importantly, dropping convexity does not violate the statistical validity: the marginal coverage guarantee holds for any measurable envelope $E \subseteq \mathbb{R}_+^K$ by exchangeability [6]. To ensure the scaling function $\tau(\mathbf{s}) = \inf\{t > 0 : \mathbf{s} \in t \cdot E\}$ yields a single, well defined scalar threshold, E needs only to be star shaped (Definition 3.2) with respect to the origin:

$$\mathbf{x} \in E \implies \lambda\mathbf{x} \in E, \qquad \forall\lambda \in [0, 1].$$

Definition 3.2 (Star-Shaped Set)**.**
A set $E \subseteq \mathbb{R}^K$ is said to be *star-shaped with respect to the origin* $0 \in \mathbb{R}^K$ if for every point $x \in E$ and every scalar $\lambda \in [0, 1]$, the line segment connecting the origin to x lies entirely within E:

$$\lambda x \in E \quad \forall x \in E,\ \forall\lambda \in [0, 1].$$

Star-shapedness guarantees that $\{t > 0 : \mathbf{s} \in t \cdot E\}$ forms a continuous interval, preserving simple single-pass calibration while allowing the envelope to tightly hug non-convex cluster geometries.

3.3 Nonconformity Score Transformation

The raw score vectors are not directly comparable across candidate labels and cannot be used directly as nonconformity measures. Recall from Section 2.4.1 that the CSA framework requires a label-dependent transformation $\phi_h : ([0, 1] \cup \{\mathtt{NA}\})^K \times \{0, 1\}^K \to (\mathbb{R}_+ \cup \{\mathtt{NA}\})^K$ that maps candidate hypotheses into a nonconformity score space centered near the origin.

The raw functional category scores $s_k \in [0, 1]$ quantify evidence in favor of functional compatibility, therefore a raw score close to 1 reflects strong infection compatibility (low nonconformity), whereas a score near 0 reflects high nonconformity. Thus, the mapping is defined such that when a candidate hypothesis $h \in \mathcal{H}$ corresponds to the true host label, the resulting vector $\mathbf{v} = \phi_h(\mathbf{s}, \mathbf{m})$ concentrates near $\mathbf{0} \in \mathbb{R}_+^K$, whereas incorrect hypotheses yield vectors situated significantly further from the origin. We now formalize ϕ_h for both classification settings while explicitly preserving structural missingness via the mask vector $\mathbf{m}$.

To map all the score vectors into one unified nonconformity space near the origin, we define label-dependent transformations ϕ_h. Formally, for a single score dimension $s_k \in [0, 1]$ with the binary mask $m_k \in \{0, 1\}$, we define the general coordinate wise mapping:

$$\phi_h(s_k, m_k) = \begin{cases} |s_k - c_h| & \text{if } m_k = 1, \\ \texttt{NA} & \text{if } m_k = 0, \end{cases} \qquad c_h \in \{0, 1\}, \tag{3.5}$$

where the c_h tells us whether a low ($c_h = 0$) or high ($c_h = 1$) raw score provides the evidence for hypothesis h. $s_k \in [0, 1]$, (3.5) can be simplified to $\phi_h(s_k) = s_k$ when $c_h = 0$, and $\phi_h(s_k) = 1 - s_k$ when $c_h = 1$, this ensures that $\phi_h(s_k) \to 0$ whenever $s_k \to c_h$.

3.3.1 Gram Type Classification Nonconformity Mapping

In the binary Gram-type classification, $\mathcal{H} = \{\text{gram-neg}, \text{gram-pos}\}$, all the K functional score models measure a single Gram-positive evidence scale. Empirically, true Gram-negative phages have low raw scores (mean $\approx 0.055 \pm 0.054$), whereas true Gram-positive phages exhibit high raw scores (mean $\approx 0.918 \pm 0.066$).

Since the scale is shared, the c_h must differ by hypothesis: $c_{\text{gramneg}} = 0$ and $c_{\text{grampos}} = 1$. This gives us the elementwise transformation for dimension $k \in \{1, \ldots, K\}$:

$$v_k = (\phi_h(\mathbf{s}, \mathbf{m}))_k = \begin{cases} s_k & \text{if } h = \text{gram-neg and } m_k = 1, \\ 1 - s_k & \text{if } h = \text{gram-pos and } m_k = 1, \\ \texttt{NA} & \text{if } m_k = 0. \end{cases} \tag{3.6}$$

Using (3.6), we get a high nonconformity score if we test an incorrect hypothesis, and the scores for the true hypothesis are mapped near the origin $\mathbf{0} \in \mathbb{R}_+^K$. This allows both classes to be calibrated against a shared reference envelope $E \subset \mathbb{R}_+^K$.

3.3.2 Host Level Classification Nonconformity Mapping

In the host level setting ($N > 2$), the candidate labels do not share the scores. Instead, each candidate host $h \in \mathcal{H}$ is evaluated on a host-specific sub-vector $\mathbf{s}_h \in ([0, 1] \cup \{\text{NA}\})^{K_h}$. This measures the direct functional compatibility with host h.

A high raw score $s_{h,k}$ indicates high likelihood of infection, $c_h = 1$ for all hosts, using this we can get the following host-specific mapping across $k \in \{1, \ldots, K_h\}$:

$$v_{h,k} = (\phi_h(\mathbf{s}_h, \mathbf{m}_h))_k = \begin{cases} 1 - s_{h,k} & \text{if } m_{h,k} = 1, \\ \text{NA} & \text{if } m_{h,k} = 0. \end{cases} \tag{3.7}$$

Unlike the Gram classification task, in host level prediction we construct N separate host envelopes $E_h \subset \mathbb{R}_+^{K_h}$ which are fitted independently on calibration phages with true label $H = h$. At test time, $\mathbf{s}_h$ is transformed using (3.7) and tested against E_h, this avoids forcing unrelated host signatures into a single shared coordinate system.

Both transforms are strictly monotone fixed maps that require no parameter estimation from data. Therefore, they do not have sample dependencies that could break the exchangeability argument (which is necessary for the coverage guarantee, Theorem 2.3). The missingness masks $\mathbf{m}$ propagate unchanged as `NA` values, and the envelope constructions of Section 3.4 operate directly on this representation.

3.4 Envelope Construction

Here we will present the three envelope geometries that we have evaluated in this report. Each method follows the two-stage calibration procedure: the shape-discovery set S_1 determines the geometric form of the envelope, and the size scaling set S_2 determines the scaling factor $\hat{t}$. All three envelopes are star-shaped with respect to the origin, so the per-point scaling function

$$\tau(\mathbf{s}) = \inf\{t > 0 : \mathbf{s} \in t \cdot E\} \tag{3.8}$$

is well-defined as a single scalar for every $\mathbf{s} \in \mathbb{R}_+^K$. For each envelope type, we derive a closed form expression for $\tau(\mathbf{s})$ that eliminates the need for iterative search. The scaling factor $\hat{t}$ is then computed using class conditional quantile aggregation over S_2, as detailed in Section 3.5.

3.4.1 Collapsed 1D Envelope Method

The simplest envelope construction method just collapses the K-dimensional (or K_h-dimensional) NC vector to a scalar value by taking the mean over the over observed dimensions,

$$\bar{s} = \frac{1}{|\mathbf{m}|_1} \sum_{k:m_k=1} v_k,$$

and calibrates a single quantile on S_1:

$$\tilde{q} = \mathrm{Quantile}_{1-\alpha}\left(\{\bar{s}_i\}_{i \in S_1}\right), \qquad \tau(\mathbf{s}) = \frac{\bar{s}}{\tilde{q}}, \qquad E = \{\mathbf{s} : \bar{s} \leq \tilde{q}\}.$$

This envelope is star-shaped (Section 3.2), it is unaffected by the dimensionality K, but at the same time is also discards all the geometric information that the K dimensions provide us with. Any two vectors with the same mean are treated identically regardless of the individual coordinates.

3.4.2 Radial Envelope Method

The Radial envelope method retains the directional structure of the scores by calibrating a boundary radius as a function of direction on the unit sphere which is restricted only to the positive orthant (since all NC values are non-negative). In this method, we draw

M directional vectors $\{\mathbf{u}_m\}_{m=1}^{M}$ uniformly over the orthant by sampling $\mathbf{v}_m \sim |\mathcal{N}(0, I_K)|$ coordinate-wise and normalizing: $\mathbf{u}_m = \mathbf{v}_m/\|\mathbf{v}_m\|$.

For each S_1 point $\mathbf{s}_i$, over the observed dimensions only, we compute its magnitude and unit direction, $r_i = \|\mathbf{s}_{i,\mathbf{m}_i}\|$, $\mathbf{d}_i = \mathbf{s}_{i,\mathbf{m}}/r_i$. For each sample direction $\mathbf{u}_m$, we collect n_m calibration points within an angular sector of half-width $\delta = 30°$ (where $\mathbf{d}_i \cdot \mathbf{u}_m \geq \cos\delta$). The directional radius $\tilde{q}_m$ is then calibrated as:

$$\tilde{q}_m = \begin{cases} \text{Quantile}_{1-\alpha}\left(\{r_i : \mathbf{d}_i \cdot \mathbf{u}_m \geq \cos\delta\}\right) & \text{if } n_m \geq n_{\min}, \\ w_m \cdot \text{Quantile}_{1-\alpha}\left(\{r_i : \mathbf{d}_i \cdot \mathbf{u}_m \geq \cos\delta\}\right) + (1-w_m)\cdot \tilde{q}_{\text{marg}} & \text{if } 0 < n_m < n_{\min}, \\ \tilde{q}_{\text{marg}} & \text{if } n_m = 0, \end{cases}$$

where n_m is the number of S_1 points within the band of direction m, $n_{\min}$ a minimum-support threshold, $w_m = n_m/n_{\min}$, and $\tilde{q}_{\text{marg}} = \text{Quantile}_{1-\alpha}(\{r_i\}_{i\in S_1})$ the marginal fallback radius. This blend enables us to avoids a hard cutoff between sufficient and insufficient local support (number of points in that region).

To ensure that we get a smooth boundary, we compute the radius at an arbitrary direction $\mathbf{d}$ as a directional weighted average of all M calibrated radii:

$$r(\mathbf{d}) = \sum_{m=1}^{M} w_m(\mathbf{d})\,\tilde{q}_m, \qquad w_m(\mathbf{d}) = \frac{\exp(\kappa\,\mathbf{d}\cdot\mathbf{u}_m)}{\sum_{m'=1}^{M}\exp(\kappa\,\mathbf{d}\cdot\mathbf{u}_{m'})}, \tag{3.9}$$

where $\kappa > 0$ controls the angular smoothness of the interpolation across directions (larger κ approaches nearest-direction assignment). This makes the envelope boundary a continuous function of direction rather than a piecewise constant one. The scaling function follows directly from this:

$$\tau(\mathbf{s}) = \frac{\|\mathbf{s_m}\|}{r(\mathbf{d})}, \qquad \mathbf{d} = \frac{\mathbf{s_m}}{\|\mathbf{s_m}\|}, \qquad E = \{\mathbf{s} : \|\mathbf{s_m}\| \leq r(\mathbf{d})\}.$$

As established in Section 3.2, this construction is star-shaped by design. This is because r depends only on direction and not on magnitude, $\mathbf{s} \in t\cdot E$ for some t implies $\lambda\mathbf{s} \in t \cdot E$ for all $\lambda \in [0,1]$.

3.4.3 Strip Envelope Method

The Strip envelope defines conditional bounds for each target dimension i based on the bin occupied by another dimension j. We partition the range of dimension j into NB equal-width bins. For each bin n containing $n_{j,n}$ calibration points, the upper limit along dimension $i \neq j$ is calibrated as:

$$\text{limits}[j,i,n] = \begin{cases} \text{Quantile}_{1-\alpha}\left(\{s_{\cdot,i} : s_{\cdot,j} \in \text{bin } n\}\right) & \text{if } n_{j,n} \geq n_{\min}, \\ w_{j,n} \cdot \text{Quantile}_{1-\alpha}\left(\{s_{\cdot,i} : s_{\cdot,j} \in \text{bin } n\}\right) + (1 - w_{j,n}) \cdot \tilde{q}_i & \text{if } 0 < n_{j,n} < n_{\min}, \\ \tilde{q}_i & \text{if } n_{j,n} = 0, \end{cases}$$

where $\tilde{q}_i$ is the marginal $(1-\alpha)$ quantile for dimension i over S_1 , and $w_{j,n} = n_{j,n}/n_{\min}$ shrinks underpopulated bins toward it (it mirrors the blending rule of the radial envelope).

A windowed monotonicity pass then enforces $\text{limits}[j,i,n] \geq \text{limits}[j,i,n']$ for $n' \in (n, n+w]$ within a fixed window w.

$$\text{limits}[j,i,n] \leftarrow \max_{n' \in [n, \min(n+w, NB-1)]} \text{limits}_{\text{raw}}[j,i,n'].$$

This is done to ensure that a bin's limit cannot be undercut by a bin immediately ahead of it in the same conditioning dimension. The monotonicity is necessary to ensure that a phage with a smaller value (more confident) in the conditioning dimension j, does not recieve a tighter bound in dimension i than a phage with a larger value in j, which makes sure that the conditional envelope is logically correct. This ensures that a single sparse or fallback-driven bin at one end of the range cannot set the limit for every bin behind it. This report uses $w \geq NB$ for gram type classification and $w = 2$ for host-level classification, reflecting the difference in how many bins (NB) each task typically calibrates.

The scaling function $\tau(\mathbf{s})$ aggregates the worst-case ratio over all conditioning pairs (j,i):

$$\tau(\mathbf{s}) = \max_{j \neq i} \frac{s_i}{\text{limits}[j,i,\text{bin}_j(\mathbf{s})]}, \qquad E = \{\mathbf{s} : \forall j \neq i,\ s_i \leq \text{limits}[j,i,\text{bin}_j(\mathbf{s})]\}.$$

Missing dimensions ($m_k = 0$) are handled pairwise:

- **Conditioning j missing, Target i present** ($m_j = 0$, $m_i = 1$): The bin index along j is undefined, so s_i falls back to its unconditional marginal ratio: $\frac{s_i}{\tilde{q}_i}$.
- **Target dimension i missing and conditioning dimension j is present** ($m_i = 0$ & $m_j = 1$): The missing target score s_i is replaced by its marginal quantile $\tilde{q}_i$, which gives us the ratio $\tilde{q}_i/\text{limits}[j,i,\text{bin}_j(s)]$.
- **Both missing:** ($m_i = 0$ & $m_j = 0$)The pair contributes no constraint (ratio $= 0$), this ensures that the unobserved dimensions do not lead to a false rejection

3.5 $\hat{t}$ Calculation

3.5.1 τ Computation

For any test point $\mathbf{s} \in \mathbb{R}^K$, we define the inclusion threshold function $\tau(\mathbf{s})$ (defined in 3.8) as the minimum scaling factor required for the scaled envelope $t \cdot E$ to contain $\mathbf{s}$:

$$\tau(\mathbf{s}) = \inf\{t > 0 : \mathbf{s} \in t \cdot E\}$$

Given the computed τ values over the scaling set S_2, the calibrated scaling factor $\hat{t}$ is selected as the $(1 - \alpha)$ empirical quantile:

$$\hat{t} = \tau_{(k)} \quad \text{where} \quad k = \lceil (|S_2| + 1)(1 - \alpha) \rceil$$

Here, $\tau_{(k)}$ denotes the k-th order statistic (the k-th smallest value) of $\{\tau(\mathbf{s}_i)\}_{\mathbf{s}_i \in S_2}$.

Unlike the original CSA method, which relies on a computationally expensive iterative binary search over candidate t values, our closed form expressions allow $\tau(\mathbf{s})$ to be evaluated directly for each point in a single pass over S_2. This eliminates the iterative search entirely, which reduces the calibration computational complexity from $O(|S_2| \cdot N_{\text{iter}})$ to $O(|S_2| \log |S_2|)$.

3.5.2 Class Specific $\hat{t}$ Calculation

When the dataset has class imbalance (like our dataset), a single global scaling factor $\hat{t}$ can cause under coverage for complex classes and over coverage for simpler ones. To enforce exact class conditional coverage $1 - \alpha$, we partition S_2 into class-specific calibration sets $S_2^{(c)} = \{\mathbf{s} \in S_2 : h = c\}$ for each label $c \in \mathcal{H}$.

For each class c, the class-specific scaling threshold $\hat{t}_c$ is calculated independently on $S_2^{(c)}$:

$$\hat{t}_c = \tau^{(c)}_{(k_c)}, \qquad k_c = \lceil (|S_2^{(c)}| + 1)(1 - \alpha) \rceil. \tag{3.10}$$

The two prediction tasks in this report differ in how envelopes are constructed, which determines how these per class thresholds $\hat{t}_c$ are applied:

Gram-Type Classification: We only fit a single envelope, E on the shared shape discovery set S_1 containing phages of both classes. If we evaluate a single pooled threshold over S_2, it would only yield only marginal coverage. So, we compute the class specific quantiles $\hat{t}_{\text{gram-neg}}$ and $\hat{t}_{\text{gram-pos}}$ using (3.10) and scale the shared envelope by their maximum:

$$\hat{t} = \max\left(\hat{t}_{\text{gram-neg}}, \hat{t}_{\text{gram-pos}}\right). \tag{3.11}$$

Since $\hat{t} \geq \hat{t}_c$ for both classes, scaling E by $\hat{t}$ satisfies both coverage requirements simultaneously, guaranteeing class-conditional validity $\mathbb{P}(\mathbf{s} \in \hat{t} \cdot E \mid H = c) \geq 1 - \alpha$ for

$c \in \{\text{gram-neg}, \text{gram-pos}\}$.

Host-Level Classification: In host-level classification, each candidate host $h \in \mathcal{H}$ has a dedicated envelope E_h fitted exclusively on phages with true host h (Section 3.1). Because its calibration set $S_2^{(h)}$ is already single-class, $\hat{t}_h$ is computed directly via 3.10 without taking a maximum across hosts. Each scaled envelope $\hat{t}_h \cdot E_h$ is validated independently, admitting a candidate host h if $\phi_h(\mathbf{s}_h) \in \hat{t}_h \cdot E_h$.

3.5.3 Resolving the Empty Prediction Set

In section 3.1 we noted that $C(\mathbf{s}, \mathbf{m})$ may be empty when no candidate label's transformed score falls inside $\hat{t} \cdot E$. In both our tasks we resolve this differently.

For gram type classification, we resolve an empty set conservatively by returning both labels, $C(\mathbf{s}) = \mathcal{H}$, rather than an arbitrary tie break, since $N = 2$, this is the uninformative but always valid choice. On the other hand, for host-level classification, where N can be large and returning all of $\hat{\mathcal{H}}(\mathbf{s})$ would be uninformative, we resolve an empty set by forcing a singleton at the host with the closest τ value:

$$C(\mathbf{s}) = \left\{ \operatorname*{argmin}_{h \in \hat{\mathcal{H}}(\mathbf{s})} \tau_h(\mathbf{s}_h) \right\}.$$

Both apply only when the primary rule of Equation (2.1) yields $\emptyset$. Coverage guarantee in Theorem 2.3 remains unaffected as nonempty fallback options strictly enlarge $C(\mathbf{s})$, thus the lower bound $\mathbb{P}(H \in C(\mathbf{s})) \geq 1 - \alpha$ trivially holds. Consequently, the reported `empty_rate` is strictly zero by rule rather than an absence of out-of-envelope points. To preserve visibility of these points, Section 5 reports the pre-resolution `forced_rate` separately.

4 Data and Preprocessing Pipeline

In this section, we describe the dataset architecture 4.1, the mathematical aggregation operators used to combine protein-level functional annotations into phage-level score vectors 4.2, and the masking strategy developed to eliminate bias caused by cross-fold protein cluster contamination 4.3.

4.1 Data Description and Structural Missingness

Our dataset comprises of $N_{\text{prot}} = 1\,853\,074$ protein sequences extracted from 18 474 unique bacteriophages. The base models use the EMPATHI pipeline (Section 2.2.2) to predict functional annotations across categories such as lysin, tail sheath, portal protein, and capsid [3]. An entry $s_{p,m} \in [0, 1]$ is the model's predicted probability that protein p carries function m. In this work we treat these scores as fixed features in our conformal framework without retraining the underlying classifiers.

Our pipeline operates on four primary data structures:

(i) **Protein Score Matrix:** Matrix of raw predictions $\mathbf{S} \in ([0, 1] \cup \{\text{NA}\})^{N_{\text{prot}} \times M}$, where entry $s_{p,m}$ denotes the predicted probability that protein p carries function m. For the Gram-type task, we evaluate the K functional categories aggregated across all hosts; for the host-level task, $N \times K = 2\,160$ host-function pairs.

(ii) **Phage Metadata:** Mappings linking each protein ID to its accession identifier, host genus label $H \in \mathcal{H}$ ($N = 54$ genera), host Gram-type $G \in \{\text{gram-neg}, \text{gram-pos}\}$, and protein cluster identifier (`pc`).

(iii) **Cross-Validation Partition:** Integer split assignments $f \in \{0, 1, 2, 3, 4\}$ assigned at the genome level, they ensure that all proteins from a given phage share the same split.

(iv) **Protein Split Mask Matrix:** A binary assignment matrix mapping each protein to its evaluation split across all host-function models.

The dataset contains some structural missingness: approximately 30.6% to 36.5% of protein category scores are missing (NA) because phages possess only a subset of functional categories. An entry $s_{p,m} = \text{NA}$ indicates that protein p was not annotated with function m. This is deterministic rather than stochastic: a capsid protein cannot carry a tail-fiber score, and an unannotated cluster does not provide any predictions.

Formally, we can say that there exists a deterministic missingness mask $\mathbf{M} \in \{0, 1\}^{N_{\text{prot}} \times M}$ such that:

$$s_{p,m} = \text{NA} \iff M_{p,m} = 0. \tag{4.1}$$

This missingness reflects the biological identity of the phage, so it is not missing randomly. Imputing NA values to zero would falsely treat unannotated proteins as evidence against infection which would violate the nonconformity interpretations of Section 3.3. We therefore preserve missingness by using NaN aware aggregation (Section 4.2).

Set	**Total Phages**	**Gram-Negative**	**Gram-Positive**
Full Dataset	18,474	10,735	7,708
Test Split (Fold 0)	4,433	2,811	1,611
Validation Split (Fold 1)	3,173	–	–
Training Splits (Folds 2–4)	10,868	7,924	6,097

Table 1: Dataset Summary and Class Distributions

4.2 Protein-to-Phage Aggregation and NaN-Aware Pooling

The base classifiers only score individual proteins $p \in \mathcal{P}_i$, while host infection is a property of the phage as a whole, so we need to aggregate the protein level scores into a single phage-level score vector. For each phage $i \in \{1, \ldots, N_{\text{phage}}\}$ ($N_{\text{phage}} = 18{,}474$) with protein set $\mathcal{P}_i$, let $M_{p,k} = \mathbf{1}(\text{protein } p \text{ has a valid prediction for column } k)$ denote the protein-level mask. We compute the aggregation using running sums and counts per phage, this avoids holding the full 1.85×10^6-row protein matrix in memory.

For each of the K functional dimensions, we get the aggregated phage level score using dimension wise, NaN aware mean pooling:

$$s_{i,k} = \begin{cases} \dfrac{\sum_{p \in \mathcal{P}_i} s_{p,k} \cdot M_{p,k}}{\sum_{p \in \mathcal{P}_i} M_{p,k}} & \text{if } \sum_{p \in \mathcal{P}_i} M_{p,k} > 0, \\ \text{NA} & \text{otherwise.} \end{cases} \tag{4.2}$$

Equation (4.2) excludes missing entries (`NA`) from both the numerator and denominator, and takes the average strictly over observed proteins. A score is `NA` if and only if no protein of phage i has a valid prediction for column k.

The phage-level mask follows directly:

$$m_{i,k} = \mathbf{1}\left(\sum_{p \in \mathcal{P}_i} M_{p,k} > 0 \right). \tag{4.3}$$

i.e. $m_{i,k} = 1$ if at least one protein of phage i has an observed score for column k, otherwise, it is 0. This reduces the protein data to a phage-level matrix $\mathbf{S} \in ([0,1] \cup \{\text{NA}\})^{N_{\text{phage}} \times K}$ paired with a binary mask for missingness $\mathbf{M} \in \{0,1\}^{N_{\text{phage}} \times K}$.

As we have already established in Section 3, our conformal envelope constructions operate natively on the paired observation $(\mathbf{s}_i, \mathbf{m}_i)$. This preserves the structural missingness without fabricating values in place of it. Consequently, all conformal classification results reported in Section 5 rely entirely on this native representation, and require no imputation after aggregation. The task dependent rules that determine which protein predictions count as valid and thus define $M_{p,k}$ are detailed in Section 4.3.

4.3 Task Specific Filtering and Data Partitioning

The main requirement for the finite sample conformal validity (Theorem 2.3) is the exchangeability of the calibration samples S_2 and test samples S_{test}. In our dataset, the cross validation partition is defined at the phage (genome cluster) level ($f_p \in \{0, 1, 2, 3, 4\}$), whereas the underlying base scoring models were trained at the protein-cluster (`pc`) level. A single protein cluster can span phages assigned to different genome folds, therefore a base model scoring protein p for category k may have observed homologous proteins from p's cluster during training.

Formally, let $F_{p,k} \in \{0, 1, 2, 3, 4\}$ denote the fold held out when training the base model for feature k. A naive aggregation which uses scores where $F_{p,k} \neq f_p$ introduces optimistic bias, by carrying training information into test evaluations. This breaks the exchangeability assumption that is a requiremenr of Theorem 2.3.

The underlying prediction models for host level and gram type classification differ in structure, hence we correct for data contamination at the appropriate granularity for each task:

For **host level classification**, each of the $M = 2\,160$ host function models was trained on an independent five fold split of the protein cluster space. `all_random_pc_splits.pkl` is the file that records $F_{p,k}$ for every protein and model. We have enforced a strict out of fold masking rule before aggregation:

$$s_{p,k} \leftarrow \text{NA} \quad \text{if } F_{p,k} \neq f_p. \tag{4.4}$$

We replace the in fold predictions with `NA`, then the NaN-aware mean pooling operator (4.2) naturally excludes the contaminated scores without introducing imputation bias. This allows us to utilize the full dataset across all five folds without discarding data:

$$D_{\text{test}} = \{i : \text{split}_i = 0\}, \qquad D_{\text{val}} = \{i : \text{split}_i = 1\}, \qquad D_{\text{train}} = \{i : \text{split}_i \in \{2, 3, 4\}\}.$$

Here, D_{train} provides us with the the shape-discovery and size-scaling calibration sets (S_1, S_2), D_{val} is used to tune envelope hyperparameters (e.g., bin count NB, directional resolution M, and kernel smoothing κ), and D_{test} is held out for final evaluation.

For **gram type classification**, the $K = 40$ functional category models used for gram type classification do not have per column split assignment tables. As a conservative, unambiguous fallback, we restrict the gram type dataset strictly to phages in Fold 0 ($\text{split}_i = 0$). Because Fold 0 phages are consistently held out at the genome cluster level, all associated protein scores are unbiased by construction. Within this Fold 0 subset, we partition the phages 40/30/30 into shape discovery (S_1), size-scaling (S_2), and test (S_{test}) sets following the standard CSA calibration split.

In both cases, $M_{p,k}$, which is the indicator determining which protein level scores are treated as valid input to the pooling operator of Equation (4.2), reduces to whichever correction each task's models support. This is the exact fold match test of Equation (4.4) for host-level classification, or membership in the Fold-0 subset for gram-type classification.

5 Experiments and Results

In this section we will present the experimental validation of the methods developed in Section 3. In section 5.1 we validate the construction of each envelope on simulated data, where the ground truth is known, we verify the finite sample coverage guarantees, geometric adaptation, and dimensionality scaling. Section 5.2 and Section 5.3 then report results on real phage data for gram-type and host-level classification respectively.

5.1 Synthetic Validation

Before applying the conformal envelope constructions of Section 3.4 to real phage data, we evaluated the theoretical guarantees, geometric adaptations and the efficiency scaling of methods on simulated scores with known ground truth.

These experiments have four specific goals:

(i) Verify the exact finite sample marginal coverage validity ($1 - \alpha = 0.900$) across varying score dimensions K.

(ii) Demonstrate how class conditional calibration improves prediction set efficiency over shared global boundaries.

(iii) Characterize how non-convex geometries (Radial and Strip) adapt to non Gaussian, correlated, or sparse score distributions.

(iv) Validate the per host envelope cascade in a multi class ground truth setting.

5.1.1 2D Geometric Proof of Concept and Class Conditional Efficiency

We first simulated $N = 400$ bivariate score pairs from $\mathcal{N}((0.15, 0.15), \Sigma)$ with $\Sigma_{11} = \Sigma_{22} = 0.01$ and $\Sigma_{12} = -0.0085$, clipped to $[0, 1]^2$. Fitting a single convex envelope ($\alpha = 0.10$, $M = 500$ directions, 50/50 S_1/S_2 split) confirms the two stage pipeline: the initial shape (only boundary under covers), whereas the size scaled envelope hits the target 90% coverage exactly.

Next, we introduce four latent classes (Hosts A–D) with centroids at $(0.1, 0.7)$, $(0.7, 0.1)$, $(0.1, 0.1)$, and $(0.7, 0.7)$, each with variance 0.01. Fitting a single shared convex envelope ($\alpha = 0.10$) to test a phage whose true host is Host B yields the prediction set $C(X) = \{\text{Host A}, \text{Host B}, \text{Host C}\}$. Because a convex hull must span the empty interior space between disjoint clusters, the prediction set remains overly conservative, even when tightening the significance level to $\alpha = 0.40$. This confirms that shared convex hull inefficiency is structural rather than an artifact of hyperparameter choice.

Computing host specific scaling thresholds $\hat{t}_h$ while sharing the directional shape $\tilde{\mathbf{q}}$ yields $\hat{t}_A = 0.969$, $\hat{t}_B = 0.980$, $\hat{t}_C = 0.302$, and $\hat{t}_D = 1.074$. The large spread in $\hat{t}_h$ reflects underlying class dispersion heterogeneity (e.g., Host C requires only 30% of the expansion required by Host D). Applying these class conditional thresholds shrinks the prediction set for the same test phage to $C(X) = \{\text{Host B}, \text{Host D}\}$, reducing set size from 3 to 2 while preserving true label coverage.

5.1.2 Dimensionality Scaling and Finite-Sample Coverage

To evaluate coverage validity as the dimensionality grows, we benchmarked the reference CSA paper method (binary search) against the Radial envelope ($\delta = 15°$), and the Strip envelope ($M_{\text{bin}} = 20$) across dimensions $K \in \{2, 5, 10, 20\}$ at target level $1 - \alpha = 0.900$ ($\alpha = 0.10, N = 1\,000$). The scores were drawn from a K-dimensional multivariate normal distribution with mean vector $\mathbf{0.3}$, variance 0.02, and off diagonal feature correlation $\rho = 0.30$, clipped to $[0, 1]^K$.

	Reference CSA (Convex)		Radial Envelope		Strip Envelope	
Dimension (K)	**Coverage (%)**	$\hat{\mathbf{t}}$	**Coverage (%)**	$\hat{\mathbf{t}}$	**Coverage (%)**	$\hat{\mathbf{t}}$
$K = 2$	90.0	0.996	90.0	0.972	90.0	0.893
$K = 5$	90.0	1.009	90.0	1.179	90.0	1.031
$K = 10$	90.0	1.016	90.0	1.920	90.0	1.107
$K = 20$	90.0	1.021	90.0	2.925	–	–

Table 2: Dimensionality scaling experiment ($K \in \{2, 5, 10, 20\}$, nominal $1 - \alpha = 0.900$, $N = 1\,000$). Empirical coverage and calibrated size-scaling factors $\hat{t}$ are reported across envelope geometries.

As shown in Table 2, every method achieves exactly 90.0% empirical coverage across all tested dimensions. This empirically confirms the distribution free guarantee of Theorem 2.3. However, their statistical efficiency, which is reflected by the scaling factor $\hat{t}$, is significantly different as K increases. For the reference CSA method, the $\hat{t}$ only slightly changes from 0.996 to 1.021 between $K = 2$ and $K = 20$. For the radial method the $\hat{t}$ value inflates for high dimensions, it nearly triples from $\hat{t} = 0.972$ at $K = 2$ to $\hat{t} = 2.925$ at $K = 20$. This is due to the curse of dimensionality, the directional cones ($\delta = 15°$) capture fewer local calibration points in sparse high dimensional spaces, so more directions fall back to the conservative global marginal quantile $\tilde{q}_{\text{marg}}$, inflating $\hat{t}$. Finally, the strip method maintains the robust sample efficiency up to $K = 10$ ($\hat{t} = 1.107$), and outperforms the radial method by using axis aligned pairwise projections. (Evaluations at $K = 20$ were omitted as per-bin sample counts shrink rapidly at fixed N).

5.1.3 Geometric Robustness Across Score Topologies

To assess the boundary adaptation under realistic non convex or misspecified feature spaces, we calibrated 3D ($K = 3$) envelopes ($\alpha = 0.10, N = 400$) across four distinct score topologies: (i) *Uncorrelated Gaussian* ($\Sigma = \mathbf{I}$), (ii) *Correlated Gaussian* ($\rho = 0.85$), (iii) *Multivariate Laplace* (heavy-tailed), and (iv) *Sparse* (one feature centered near zero, 0.05 ± 0.03, while others remain diffuse at 0.35).

Table 3: Geometric adaptation benchmark across 3D score topologies ($\alpha = 0.10, N = 400$). Both Radial ($M = 150$) and Strip ($M_{\text{bin}} = 12$) envelopes achieve exact nominal coverage under distributional misspecification.

	Radial Envelope		**Strip Envelope**	
Score Distribution Topology	**Coverage (%)**	$\hat{\mathbf{t}}$	**Coverage (%)**	$\hat{\mathbf{t}}$
Normal (Uncorrelated, $\Sigma = \mathbf{I}$)	90.0	1.072	90.0	1.161
Normal (Highly Correlated, $\rho = 0.85$)	90.0	0.995	90.0	0.861
Multivariate Laplace (Heavy-Tailed)	90.0	1.072	90.0	1.232
Sparse (Degenerate Dimension)	90.0	1.077	90.0	1.155

The results in Table 3 highlight several key properties:

- **Distribution Free Invariance:** Both Radial and Strip constructions hit the exact 90.0% target across all four topologies. this confirms that heavy tails or feature sparsity do not compromise marginal validity.
- **Adaptation to Correlation:** On highly correlated data ($\rho = 0.85$), both methods require little to no inflation ($\hat{t} \approx 1$ for radial and $\hat{t} \approx 0.861$ for strip). This indicates that the shape discovery step already produces a well fitted envelope for diagonal score clouds. The Strip method's initial conditional bounds are slightly conservative for this geometry, so the calibration step fine tunes the scale downward.
- **Heavy-Tail Sensitivity:** The heavy tailed Laplace distribution induces the largest inflation factor for the Strip envelope ($\hat{t} = 1.232$). This is because heavy tails produce extreme nonconformity values that force wider bin wise quantile limits during shape discovery.

5.1.4 Multi-Host Benchmark Verification

Finally, we validate the per-host envelope cascade (Section 3.1) on a simulated 6-host system ($\mathcal{H} = \{\text{Host}_A, \ldots, \text{Host}_F\}$, $K = 5$, $N_{\text{cal}} = 500$ infectors per host, $\alpha = 0.10$). Host-specific envelopes are calibrated from true infector score distributions centered in $[0.1, 0.5]^5$. A test phage matching Host_B (scores in $[0.1, 0.4]^5$ for Host_B; scores in $[0.6, 0.9]^5$ for all non target hosts) is evaluated against all six envelopes.

Across the reference CSA, Radial, and Strip constructions, all three methods correctly isolate $C(\mathbf{s}) = \{\text{Host}_B\}$ as an ideal singleton prediction set, excluding every non-target host. This test case serves as a sanity check under separated ground truth rather than a full coverage estimate, but it also confirms the per host cascade functions as intended prior to evaluating the substantially larger host space ($N = 54$) in Section 5.3.

5.2 Gram Type Classification

We evaluated the three envelope geometries (Radial, Strip, and Collapsed 1D) on the real phage data for a binary Gram type classification ($\mathcal{H} = \{\text{gram-neg}, \text{gram-pos}\}$). We only used the 4433 phages in split 0 to ensure that all scores remain out of fold by construction and there is no contamination.

5.2.1 Experimental Setup

We partitioned the split 0 subset into shape discovery ($|S_1| = 1\,768$), size scaling ($|S_2| = 1\,326$), and test evaluation ($|S_{\text{test}}| = 1\,328$) sets using a 40/30/30 split. For each geometry, we fitted a single shared envelope E on S_1, and then calculated the class conditional scaling factors $\hat{t}_{\text{neg}}$ and $\hat{t}_{\text{pos}}$ on S_2 using (3.10). To ensure validity across both classes simultaneously, we scaled the envelope by their maximum:

$$\hat{t} = \max\left(\hat{t}_{\text{neg}},\, \hat{t}_{\text{pos}}\right). \tag{5.1}$$

At test time, we evaluated the Gram type nonconformity transformation under both candidate hypotheses. For an empty candidate set ($\emptyset$), we returned both labels ($C(\mathbf{s}) = \mathcal{H}$) as established in Section 3.5.

5.2.2 Coverage and Prediction Efficiency

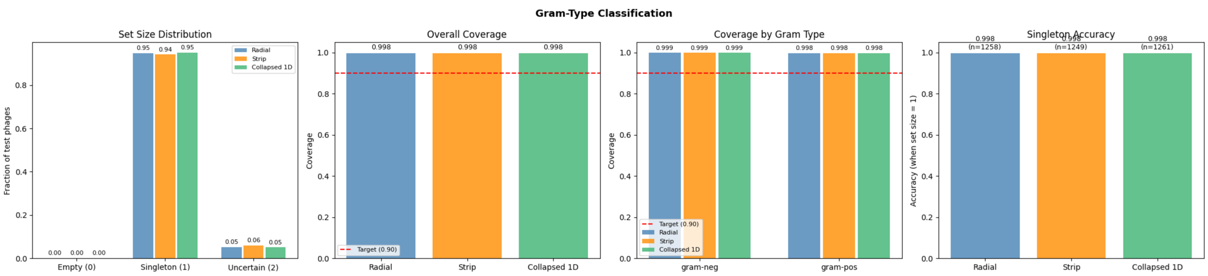

Figure 1: Gram type classification performance across Radial, Strip, and Collapsed 1D geometries on the test split (Fold 0, nominal target $1 - \alpha = 0.90$). Panels show (left to right): prediction set size distribution, marginal empirical coverage, class-conditional coverage, and singleton accuracy with sample counts (n).

As we can see in Figure 1 (middle panels), all three envelope geometries achieve 99.8% empirical coverage, which is well above the nominal 90% target (red dashed line). The class conditional breakdown confirms that the coverage is balanced for both Gram negative

(99.9%) and Gram positive (99.8%) phages. This demonstrates that the max threshold calibration $\hat{t} = \max(\hat{t}_{\text{neg}}, \hat{t}_{\text{pos}})$ effectively eliminates coverage disparity between classes.

We can also see that the high coverage does not compromise efficiency. The set size distribution chart of Figure 1 (left panel), shows that 94–95% of test phages receive singleton predictions, while only 5–6% require size 2 uncertain sets ($C(\mathbf{s}) = \mathcal{H}$). The empty set rate is 0% by construction due to the conservative fallback rule. When singleton sets are produced, the prediction accuracy is 99.8% across all three geometries ($n \geq 1\,249$), which tells us that when the model is confident enough to output exactly one label, it is almost always correct.

Acheiving this coverage that is over the target highlights a trade off, it suggests the calibrated envelopes are wider than strictly necessary to meet $1 - \alpha = 0.90$ (due to the max threshold calibration). So we are trading efficiency for a safety margin the task may not require. We will return to this trade off in Section 6.

5.2.3 Score Separation and Geometric Equivalence

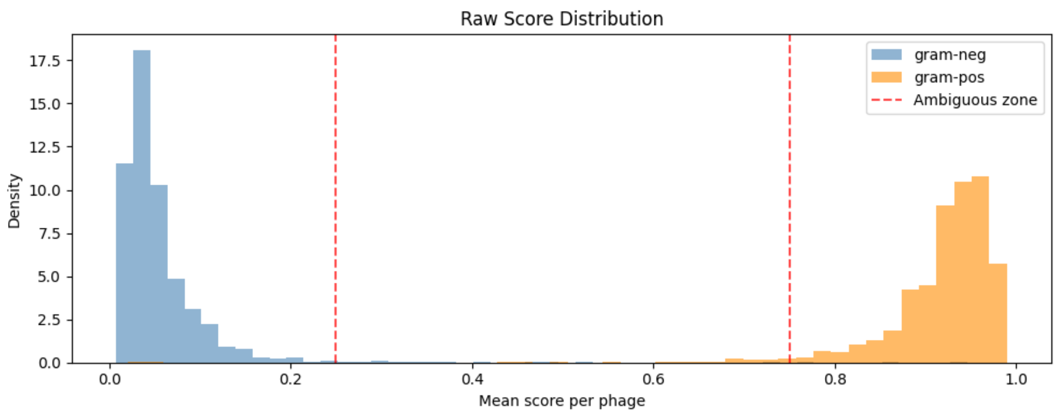

Figure 2: Distribution of raw scores per test phage, separated by true Gram type. Gram negative phages concentrate tightly near 0 while Gram positive phages concentrate near 1, leaving the intermediate ambiguous zone (0.25–0.75, dashed red lines) almost empty.

All three envelope constructions yield nearly identical performance metrics in Figure 1. This equivalence can directly be explained by the empirical raw score distribution in Figure 2: Gram negative phages concentrate tightly near the origin (mean 0.055 ± 0.054), whereas Gram positive phages concentrate near 1.0 (mean 0.918 ± 0.066). The two classes form disjoint, well separated clusters with negligible density in the ambiguous zone (0.25–0.75), any sensible star shaped boundary cleanly separates them.

This near identical performance across the three geometries suggests that the choice of envelope geometry is less critical for low dimensional task with strongly separated scores. The advantages of non convex can be seen more prominently in the higher dimensional, multi class host level classification task in Section 5.3

5.3 Host Level Classification

Host level prediction extends our conformal prediction framework from binary attributes to a multiclass task across $N = 54$ candidate bacterial hosts ($\mathcal{H} = \{h_1, \ldots, h_54\}$). This setting introduces statistical and dimensional challenges: heavy label imbalance, small per host calibration samples, and inference through a hierarchical two-stage cascade. Rather than attempting to fit a 2 036 dimensional global envelope across the joint feature space, we constructed class conditional envelopes which were calibrated independently for each candidate host $h \in \mathcal{H}$ on its dedicated K_h dimensional score sub vectors $\mathbf{s}_h$.

5.3.1 Experimental Protocol

We apply the split conformal framework of Section 3 to the reconstructed score vectors (3.7) of the $N_{\text{tot}} = 15{,}638$ annotated phages across 54 candidate hosts ($K_h \in [26, 40]$ per host; $K_{\text{tot}} = 2{,}036$). We again use a three way partition of the genome cluster data:

- D_{train} (Split 2–4, $N_{\text{train}} = 9{,}101$): For each candidate host with sufficient training data ($n_h \geq 3$), the infector phages are partitioned 50/50 into shape-discovery ($S_1^{(h)}$, $\lfloor n_h/2 \rfloor$) and size-scaling ($S_2^{(h)}$, $\lceil n_h/2 \rceil$) subsets to fit E_h and compute $\hat{t}_h$.
- D_{val} (Split 1, $N_{\text{val}} = 2{,}593$): used for envelope hyperparameter tuning.
- D_{test} (Split 0, $N_{\text{test}} = 3{,}944$): held out and evaluated once after fixing all parameters.

Unlike Gram type classification (Section 5.2), host level scores use per model split assignments (Section 4.3) to ensure the out of split validity across all five splits rather than restricting calibration to split 0 alone. Missing entries (NA) are handled by each envelope geometry without data imputation

At test time, the Gram restriction $\hat{\mathcal{H}}(\mathbf{s})$ narrows the host space to 16 for Gram positive and 38 for Gram negative predictions. The cascaded prediction set $C_{\text{host}}(\mathbf{s})$ is evaluated using the per host inclusion rule of Equation (3.1). The cascade coverage guarantee of Proposition 3.1 applies directly, bounding theoretical miscoverage by $\leq 2\alpha = 0.20$. If a test phage falls outside the calibrated envelopes of all candidate hosts ($C_{\text{host}}(\mathbf{s}) = \emptyset$), the empty set is resolved by selecting the candidate minimizing normalized nonconformity:

$$h_{\text{forced}}(\mathbf{s}) = \underset{h \in \hat{\mathcal{H}}(\mathbf{s})}{\arg\min} \frac{\tau_h(\mathbf{s}_h)}{\hat{t}_h}, \tag{5.2}$$

which normalizes distances across different host geometries without defaulting to the uninformative full label set.

5.3.2 Hyperparameter Selection on Validation Data

We optimized the hyperparameters on D_{val} (Split 1) by minimizing the average set size subject to empirical coverage constraints at target level $1-\alpha = 0.90$::

$$\theta^\star \in \arg\min_{\theta} \overline{|C|}_{\text{val}}(\theta) \quad \text{subject to} \quad \hat{c}_{\text{val}}(\theta) \geq 1-\alpha. \tag{5.3}$$

with ties or near constraint cases resolved by minimizing the coverage discrepancy $|\hat{c}_{\text{val}}(\theta) - (1-\alpha)|$.

Validation under the unrestricted candidate space selected:

- **Strip Binning (N_B):** Swept over $N_B \in \{3, 5, 8, 10, 15, 20\}$ with windowed monotonic smoothing ($w = 2$). $N_B = 8$ was selected, achieving 90.2% validation coverage with an average prediction set size of 13.43 candidate hosts.

- **Radial Resolution (M, κ):** Swept across $M \in \{100, 250, 500\}$ directions and concentration parameters $\kappa \in \{4.0, 8.0, 16.0\}$. $M = 250$ with $\kappa = 4.0$ provided the optimal trade-off closest to nominal validity (89.0% validation coverage, average set size 12.75).

5.3.3 Empirical Test Results (Split 0)

Table 4 summarizes performance metrics evaluated on the test split ($N_{\text{test}} = 3{,}944$ phages) across all 54 hosts.

Method	Coverage	Avg. Set Size	Singleton Rate	Singleton Acc.	Forced Rate
Radial Envelope ($M = 250, \kappa = 4$)	0.933	15.74	0.066	0.762	0.016
Strip Envelope ($N_B = 8, w = 2$)	0.904	15.55	0.001	–	0.000
Collapsed 1D	0.927	7.11	0.166	0.907	0.006

Table 4: Performance metrics on the test split

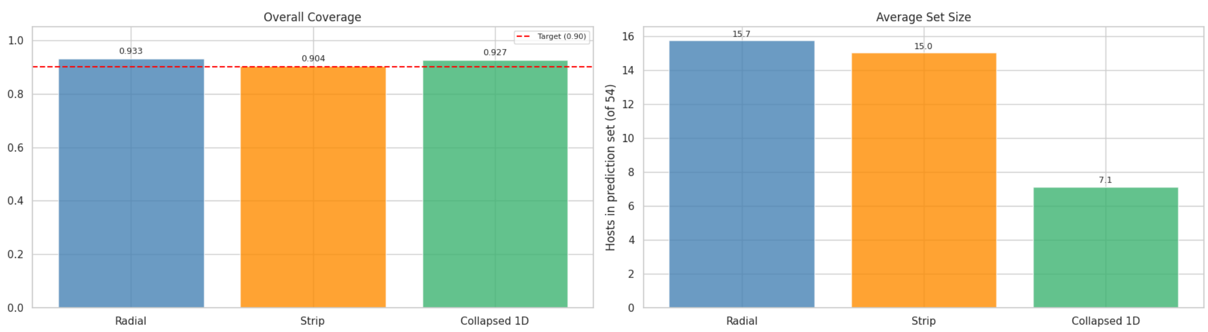

Figure 3: Host classification performance on Split 0 test phages ($1-\alpha = 0.90$). Left panel: overall empirical coverage compared against the target (red dashed line). Right panel: average prediction set size across candidate hosts.

As demonstrated in Table 4 and Figure 3, all three geometries achieve empirical coverage close to the nominal 90.0% target. In contrast to the binary Gram setting where max threshold calibration yielded 99.8% coverage, independent per host calibration eliminates artificial over coverage.

Prediction efficiency diverges sharply:

(i) **Efficiency of Collapsed 1D:** Collapsed 1D achieves an average prediction set size of 7.11, which is less than half that of Radial (15.74) or Strip (15.55). In moderate to high dimensions ($K_h \geq 26$), 1D mean pooling avoids the sample sparsity penalties that arise due to angular cones and multi dimensional bins. The complete per host set size distributions and boxplots across all 54 hosts are in Appendix A Figure 5 and 6

(ii) **Singleton Accuracy:** Collapsed 1D isolates a single host for 16.6% of test phages and achieves 90.7% accuracy on these singletons. In contrast, Radial produces singletons for only 6.6% of test phages (with only 76.2% accuracy), while Strip almost never gives a singleton output (0.1%).

(iii) **Forced Decision Rate:** The forced fallback rule (5.2) was only used for 0.6% of test phages for Collapsed 1D and 1.6% for Radial. This confirms that host specific envelopes reliably capture empirical infector score distributions.

Figure 4 in the Appendix A illustrates the class conditional validity across all hosts. The coverage remains well calibrated across major host types (e.g., *Escherichia*, *Pseudomonas*, *Staphylococcus*, and *Mycobacterium*), but we can also find low coverage to the host classes with sparse data.

6 Conclusion and Future Work

6.1 Summary.....

6.2 Limitations

- gram-pos coverage -

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A Host Classification Results

A.1 Class Conditional Coverage Breakdown

A.2 Set Size Distribution Per Host

A.3 Prediction Set Inclusion Heatmaps

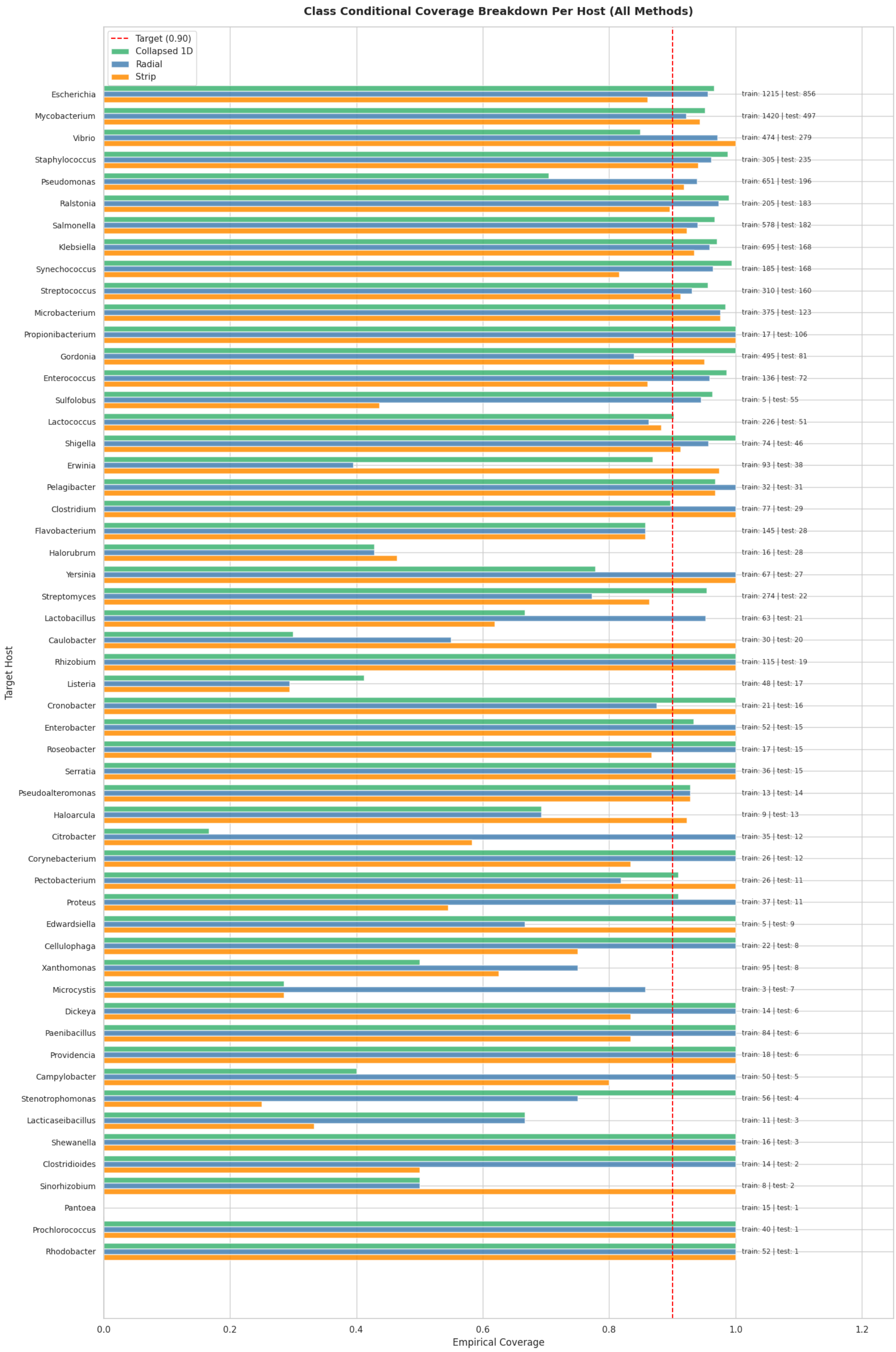

Figure 4: Class conditional empirical coverage across all 54 candidate bacterial hosts on Split 0 test phages. The red dashed line denotes the nominal $1-\alpha = 0.90$ target.

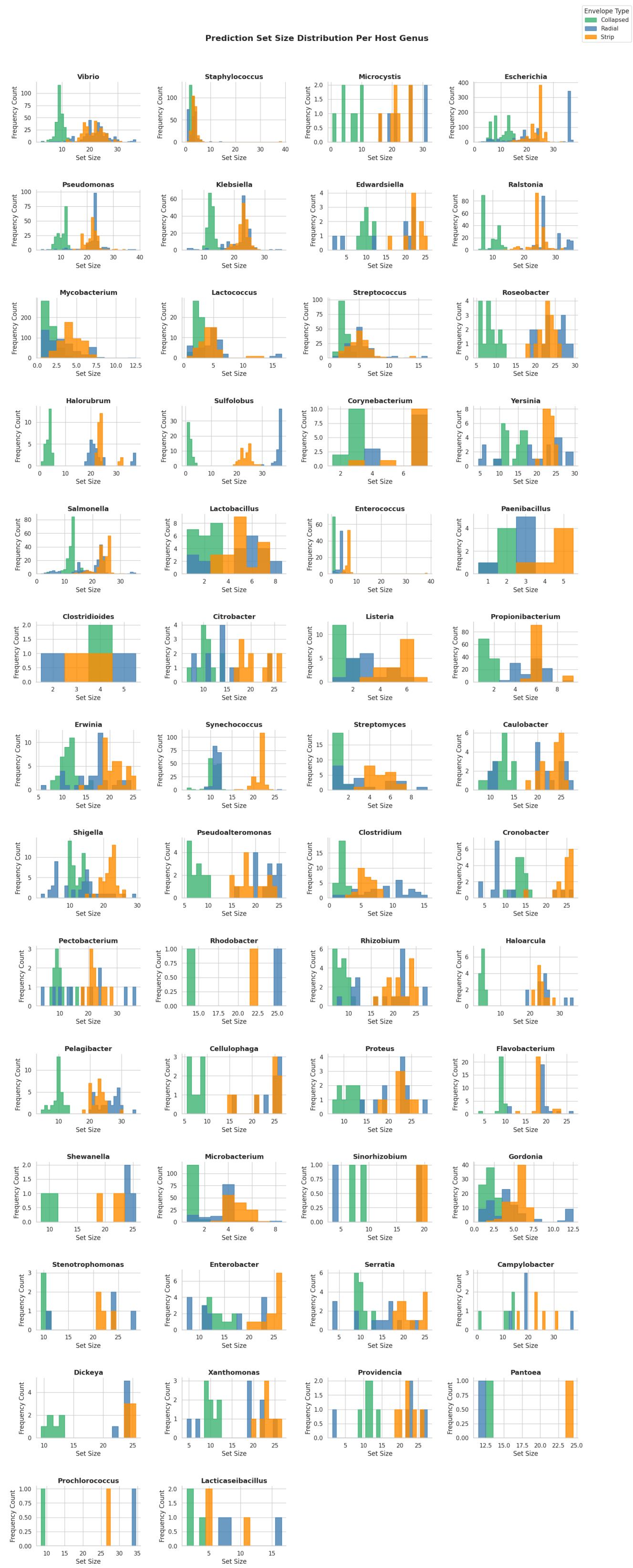

Figure 5: Prediction set size histograms for all 54 host types across Collapsed 1D, Radial, and Strip envelopes.

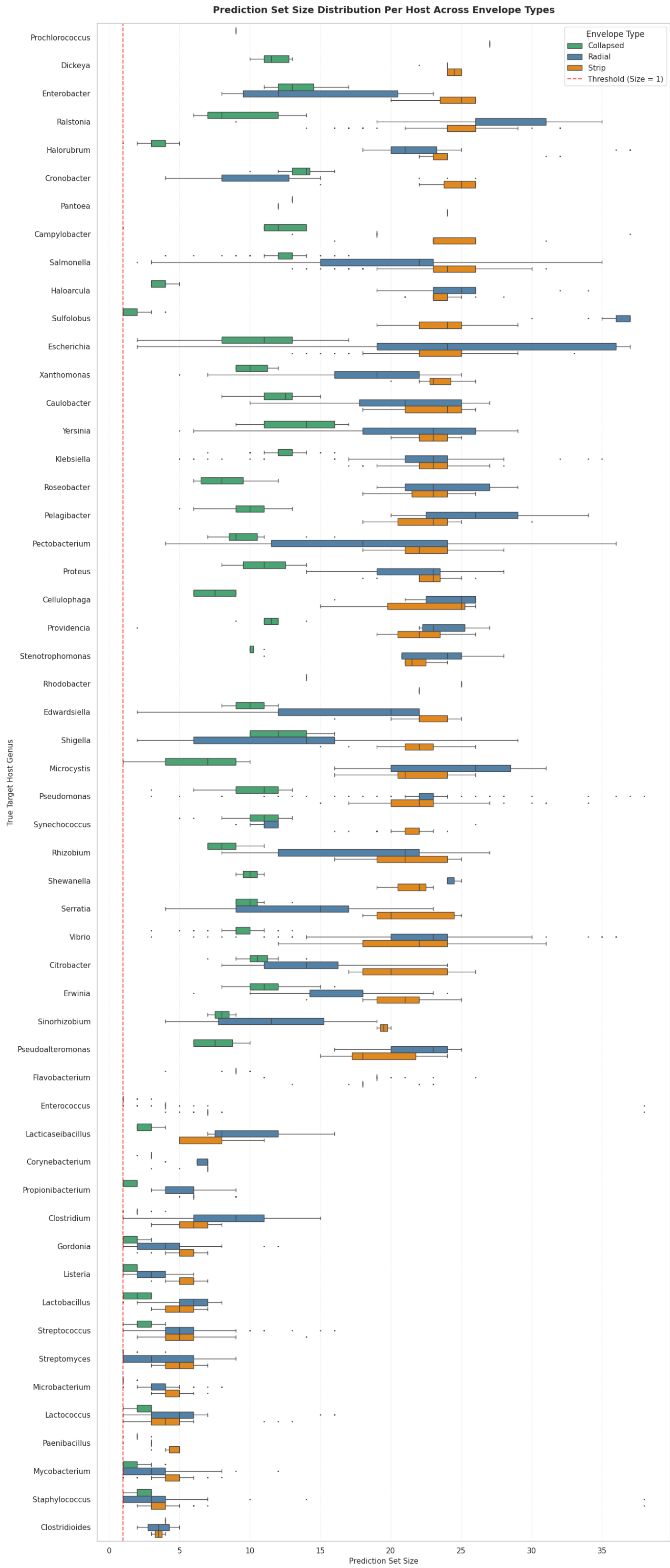

Figure 6: Per host prediction set size distribution across envelope geometries, ordered by training set size.

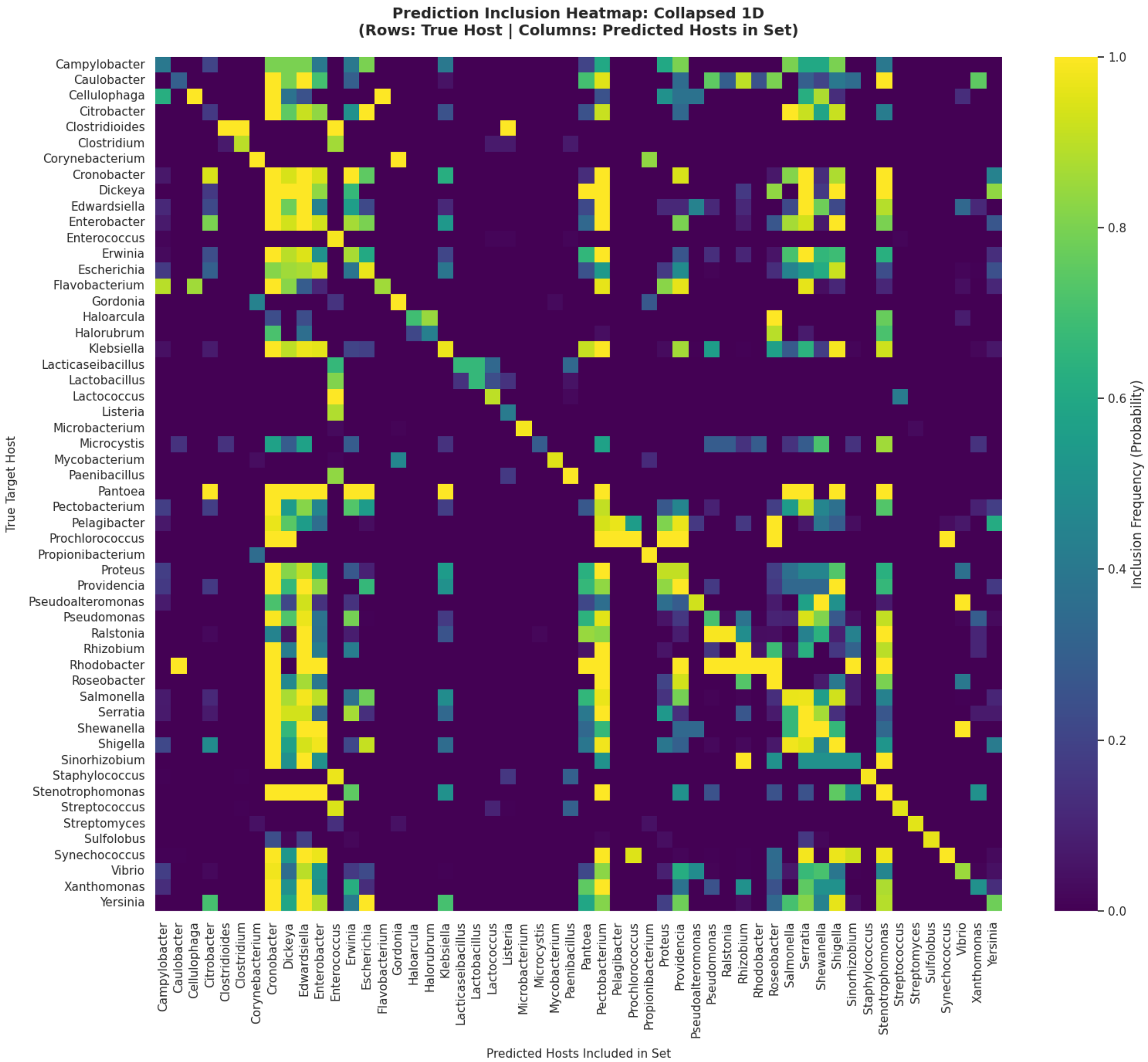

Figure 7: Prediction set inclusion frequency heatmap for the 1D envelope across all 54 true hosts (rows) vs. predicted candidate hosts (columns).

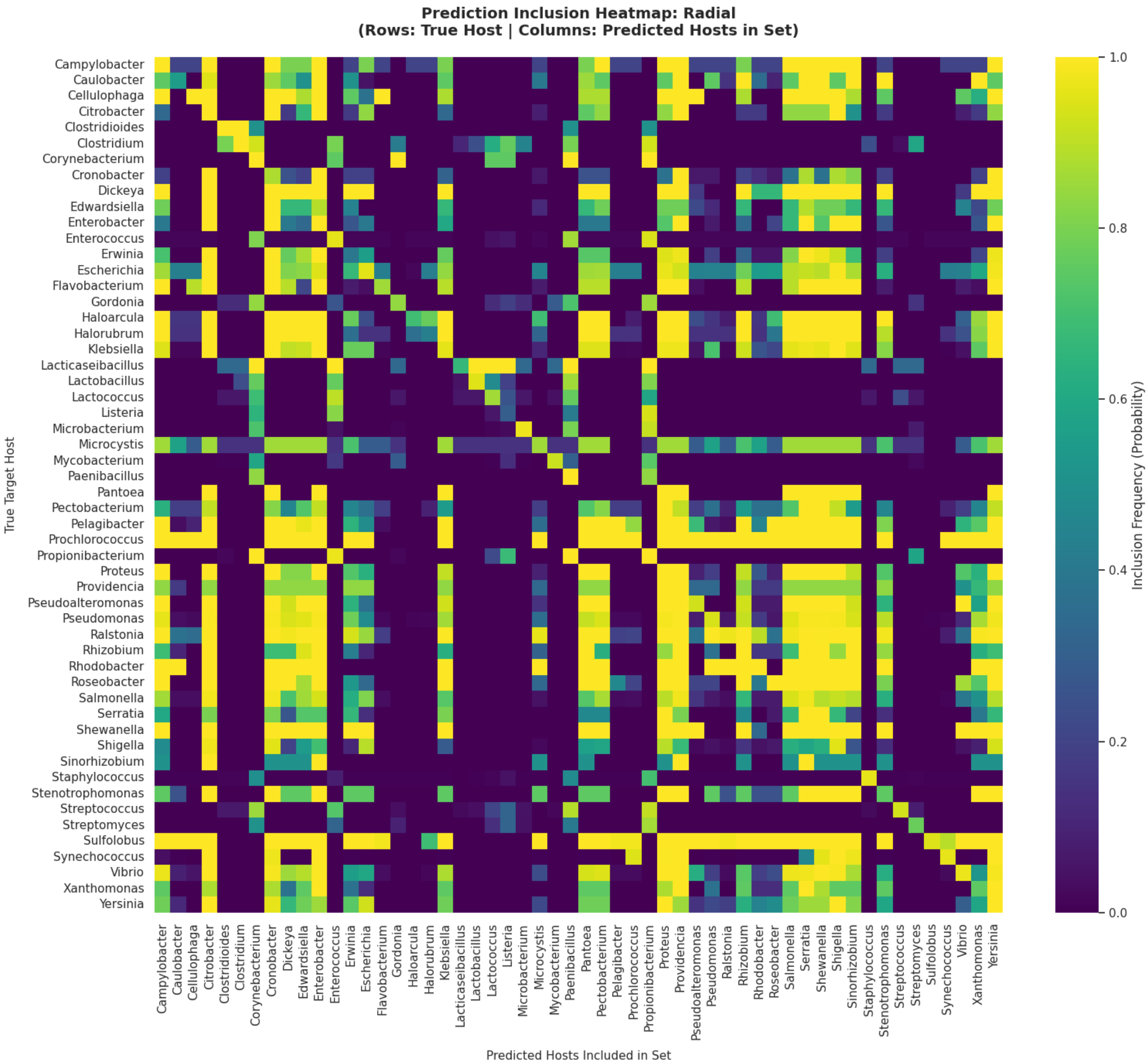

Figure 8: Prediction set inclusion frequency heatmap for the Radial envelope across all 54 true hosts (rows) vs. predicted candidate hosts (columns).

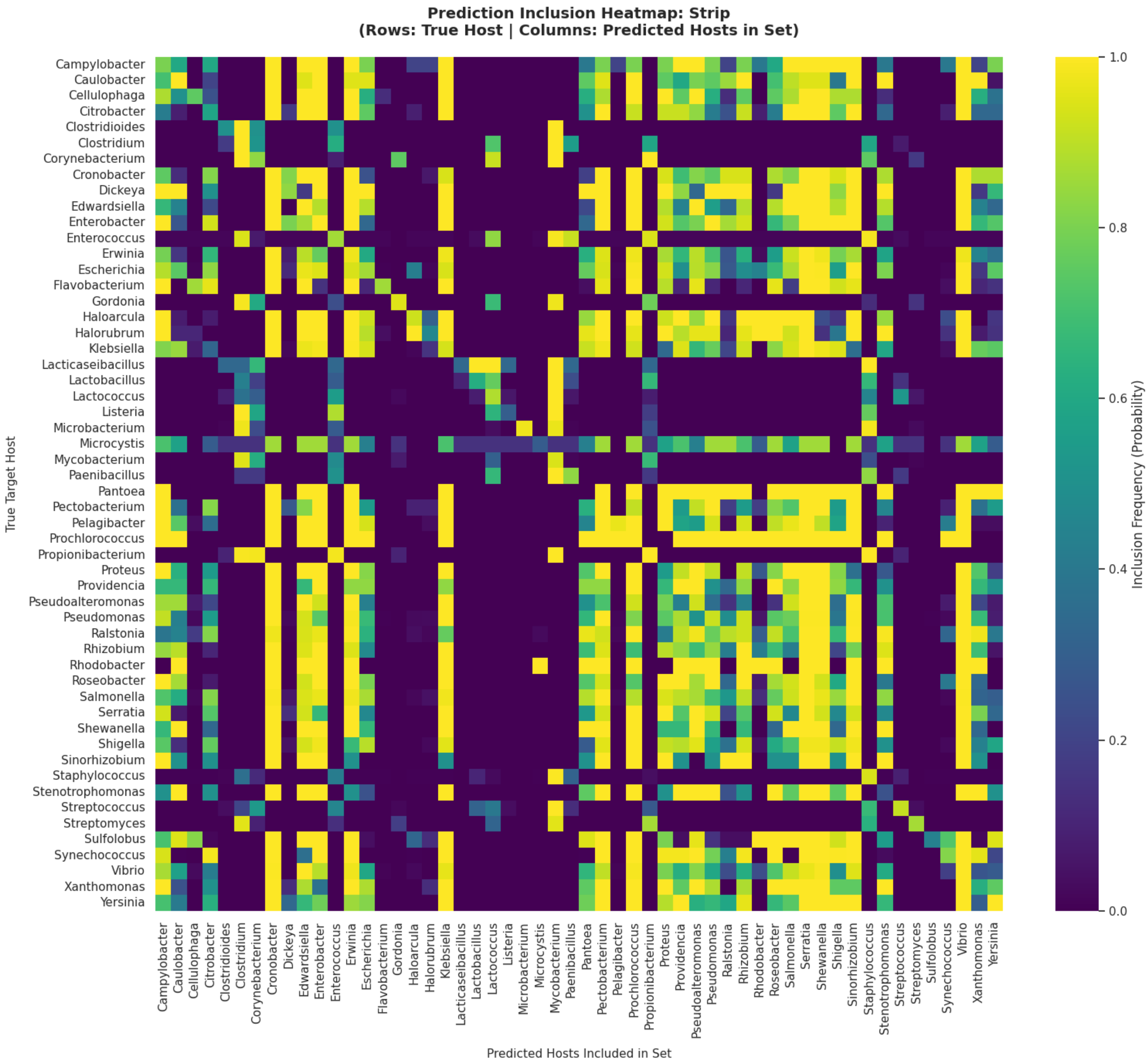

Figure 9: Prediction set inclusion frequency heatmap for the Strip envelope across all 54 true hosts (rows) vs. predicted candidate hosts (columns).